

Implementation and Verification of a Two-Temperature Model for Hypersonic Flows

Research Internship

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Nomenclature

Symbols

α	Thermal diffusivity [$\text{m}^2 \text{s}^{-1}$]	$\mathbf{u}$	velocity vector [m/s]
α	Vibrational quantum level index [-]	μ	Mixture dynamic viscosity [$\text{kg m}^{-1} \text{s}^{-1}$]
$\alpha_{s,r}$	Stoichiometric coefficient of species s as reactant in reaction r [-]	μ_s	Dynamic viscosity of species s [$\text{kg m}^{-1} \text{s}^{-1}$]
$\bar{c}_m$	Mean molecular speed of species m [m s^{-1}]	μ_{mj}	Reduced molecular weight of species pair (m, j) [kg mol^{-1}]
$\bar{T}$	reference temperature used in polynomial evaluation [K]	ϕ_s	Wilke mixing function for species s [-]
$\bar{E}(T_{tr}, T_{tr})$	Mean vibrational energy for backward (recombination) process [J]	$\pi \bar{\Omega}_{sr}^{(1,1)}$	Collision integral of order (1,1) for species pair (s, r) [m^2]
$\bar{E}(T_{tr}, T_{ve,m})$	Mean vibrational energy for forward (dissociation) process [J]	$\pi \bar{\Omega}_{sr}^{(2,2)}$	Collision integral of order (2,2) for species pair (s, r) [m^2]
β_r	Temperature exponent in Arrhenius law for reaction r [-]	$\Re_m$	Maximum vibrational level (cut-off before dissociation) for species m [-]
$\beta_{s,r}$	Stoichiometric coefficient of species s as product in reaction r [-]	ρ	mixture mass density [kg/m^3]
$\Delta_{sr}^{(1)}$	Modified collision integral of order (1) for species pair (s, r) [$\text{m}^3 \text{s}^{-1}$]	ρ_s	mass density of species s [kg/m^3]
$\Delta_{sr}^{(2)}$	Modified collision integral of order (2) for species pair (s, r) [$\text{m}^3 \text{s}^{-1}$]	$\sigma^{Casseau}$	Reference collision cross-section in Casseau model [m^2]
$\dot{\omega}_s$	Mass production rate of species s [$\text{kg m}^{-3} \text{s}^{-1}$]	σ^{Gnoffo}	Constant collision cross-section in Gnoffo model [m^2]
η	Mixture thermal conductivity [$\text{W m}^{-1} \text{K}^{-1}$]	$\sigma^{Scalabrin}$	Reference collision cross-section in Scalabrin model [m^2]
η_q	Mixture thermal conductivity for energy mode q [$\text{W m}^{-1} \text{K}^{-1}$]	σ_m	Effective collision cross-section for species m [m^2]
$\eta_{q,s}$	Thermal conductivity of species s for energy mode q [$\text{W m}^{-1} \text{K}^{-1}$]	τ_m	Vibrational-translational relaxation time of species m [s]
$\eta_{V,s}$	Vibroelectronic thermal conductivity of species s [$\text{W m}^{-1} \text{K}^{-1}$]	τ_m^{MW}	Millikan-White relaxation time [s]
η_V	Mixture vibroelectronic thermal conductivity [$\text{W m}^{-1} \text{K}^{-1}$]	τ_m^P	Park high-temperature correction to relaxation time [s]
$\hat{D}_m$	Effective dissociation energy for species m including electronic contribution [J kg^{-1}]	$\theta_{el,i,s}$	characteristic electronic temperature of level i of species s [K]
		$\theta_{v,m}$	Characteristic vibrational temperature of species m [K]
		$\theta_{v,s}$	characteristic vibrational temperature of species s [K]
		$\varepsilon_{v,\alpha,m}$	Energy of vibrational level α of species m [J]
		$\zeta_{el,s}$	Electronic degrees of freedom of species s [-]

$\zeta_{r,s}$	Rotational degrees of freedom of species s [-]	E	total energy per unit mass of mixture [J/kg]
$\zeta_{t,s}$	Translational degrees of freedom of species s [-]	e_s	total energy per unit mass of species s [J/kg]
$\zeta_{V,s}$	Vibroelectronic degrees of freedom of species s [-]	e_V	mixture vibrational energy per unit mass [J/kg]
$\zeta_{v,s}$	Vibrational degrees of freedom of species s [-]	$e_{el,s}$	electronic energy per unit mass of species s [J/kg]
A	Pre-exponential factor in Arrhenius law	$e_{r,s}$	rotational energy per unit mass of species s [J/kg]
A_m	Millikan–White empirical coefficient for species m [-]	$e_{t,s}$	translational energy per unit mass of species s [J/kg]
A_r	Pre-exponential factor in Arrhenius law for reaction r (units depend on reaction order)	$e_{V,s}$	vibroelectronic internal energy per unit mass of species s evaluated at temperature T_V [J/kg]
A_s, B_s, C_s	Blottner curve-fit coefficients for species viscosity [-]	$e_{v,s}$	vibrational energy per unit mass of species s [J/kg]
a_{sr}	Interaction coefficient for species pair (s, r) [-]	$e_{V,s}^*$	vibroelectronic internal energy per unit mass of species s evaluated at temperature T [J/kg]
C_p	Mixture heat capacity at constant pressure [J kg ⁻¹ K ⁻¹]	$g_{i,s}$	degeneracy of electronic level i of species s
$C_{p,r}^s$	rotational specific heat at constant pressure of species s [J/(kg K)]	h_s	transrotational enthalpy per unit mass of species s [J/kg]
$C_{p,t}^s$	translational specific heat at constant pressure of species s [J/(kg K)]	h_V	vibroelectronic enthalpy per unit mass of species s [J/kg]
c_{pref}	Preferential dissociation coefficient [-]	$h_{s,0}$	reference enthalpy of species s at T_{ref} [J/kg]
$C_{v,el,s}$	Electronic heat capacity at constant volume of species s [J kg ⁻¹ K ⁻¹]	$h_{V,s}$	vibroelectronic enthalpy per unit mass of species s [J/kg]
$C_{v,V,s}$	Vibroelectronic heat capacity at constant volume of species s [J kg ⁻¹ K ⁻¹]	k_B	Boltzmann constant [J K ⁻¹]
$C_{v,v,s}$	Vibrational heat capacity at constant volume of species s [J kg ⁻¹ K ⁻¹]	k_f	Forward reaction rate constant
D'_m	Modified vibrational contribution to dissociation energy for species m [J kg ⁻¹]	$k_{b,r}$	Backward reaction rate constant for reaction r
D_m	Dissociation energy of species m [J kg ⁻¹]	K_{eq}	Equilibrium constant of reaction r [-]
D_s	Effective diffusion coefficient of species s [m ² s ⁻¹]	$k_{f,r}$	Forward reaction rate constant for reaction r
D_{sr}	Binary diffusion coefficient between species s and r [m ² s ⁻¹]	Le	Lewis number [-]
		M_s	molecular weight of species s [kg/mol]

m_s	Mass of one particle of species s [kg]	T_{tr}	Translational-rotational temperature [K]
n_j	Number density of collision partner species j [m^{-3}]	$T_{ve,m}$	Vibroelectronic temperature of species m [K]
n_m	Number density of species m [m^{-3}]	U_m	Marrone–Treanor parameter for species m [K]
N_r	Number of chemical reactions [-]	y_s	mole fraction of species s
p	pressure [Pa]	$Z(T)$	Vibrational partition function [-]
p_s	partial pressure of species s [Pa]	$\bar{R}$	universal gas constant [$\text{J}/(\text{kg mol K})$]
$Q_{m,V-T}$	Vibrational–translational energy exchange source term for species m [W m^{-3}]	τ_s^{MW}	translational-vibrational relaxation time constant for species s from correlation of Millikan and White [s]
R_s	specific gas constant of species s , $\text{J}/(\text{kg K})$	$\langle \tau_s \rangle$	translational-vibrational energy relaxation time constant for species s [s]
$R_{b,r}$	Backward reaction rate of reaction r [$\text{mol m}^{-3} \text{s}^{-1}$]	$\hat{D}_s$	average vibrational energy per unit mass of species s , which is created or destroyed at mass production rate $\dot{\omega}_s$ [J/kg]
$R_{f,r}$	Forward reaction rate of reaction r [$\text{mol m}^{-3} \text{s}^{-1}$]	Subscripts	
T	transrotational temperature [K]	i	index of electronic energy levels
T_a	Activation temperature [K]	j	Index of collision partner species
T_V	vibroelectronic temperature [K]	e	electronic energy mode
T_{ref}	reference temperature [K]	m	molecules i.e. polyatomic species m
$T_{a,r}$	Activation temperature of reaction r [K]	r	rotational energy mode
$T_{c,b}$	Rate-controlling temperature for backward reaction [K]	s	species s
$T_{c,f}$	Rate-controlling temperature for forward reaction [K]	t	translational energy mode
$T_{F,m}$	Modified (Treanor) temperature for species m [K]	V	vibroelectronic energy modes of Two-Temperature model
T_{ov}	Overall (effective) temperature combining all energy modes [K]	v	vibrational energy mode

1 Introduction

This semester thesis presents the results from a hypersonic CFD solver accelerated by the Python library *JAX* which promises accelerator-oriented array computation [1]. Most interesting for this project are the *Just-In-Time* compilation and auto-vectorization capabilities of JAX. With the right integration, a more complex problem will take the same amount of time to compute compared to an easier as the complexity is handled by more compute resources. Hypersonic regimes often imply low fluid densities. This can impose challenges for computational fluid dynamic (CFD) solvers as the densities might be so low that the continuum assumption is violated which is required to be valid to ensure the applicability of the Navier-Stokes equations. An extension of the physical correctness can be achieved by considering effects which can happen on much shorter time scales than the convective flow. Two effects of this type will be studied closer in the scope of this work. First, energy is allowed to be in in-equilibrium across the different energy modes of a molecule. Different quantization of these energy modes leads to an uneven excitation if a strong force like the compression of a shock wave hits the molecule. This is relevant for the second effect which can occur on much smaller time scales compared to the energetic or thermal equilibrium, which is that chemical reactions occur with a finite rate instead of infinitely fast. Chemical inequilibrium assumes that it takes a finite amount of time for a species to reach chemical equilibrium meaning until dissociation or recombination stopped, after energy was added to the system. In the scope of this work, the thermal inequilibrium is modeled with a Two-Temperature model which categorizes the energy modes into two different classes according to their quantization. The chemical inequilibrium is model via reaction-individual Arrhenius equations.

This work starts by discussing the different approaches to model hypersonic fluids on a high level (cf. Section 2). Section 3 presents the different equation sets which have been implemented in code. An overview about the numerics and the most important data types is presented in Section 4.2 and 4.1. Section 5 finally presents results in 0D, 1D and 2D and compares against results from literature.

2 Computational Methods

The following sections represent the main orthogonal modeling dimensions to simulate hyper-sonic flows. The scenario to be modeled, fidelity requirements and available resources are examples for criteria that determine the right selection of physic models.

2.1 Fluid Description

A fluid can be described at different levels of fidelity. One of the highest level of fidelity is a description with single particles, and on the lower end of fidelity is a continuum assumption which blurs the particle picture into bulk motion. The following section explains the connection between the two to set the assumptions made by different levels of fidelity into context.

2.1.1 The Boltzmann Equation

The Boltzmann equation represents the velocity distribution of all considered particles in phase space and, thus, models a fluid at molecular scale. The equation depends on the velocity vector, the position vector as well as other degrees of freedom the fluid's particles possess. According to Boyd [2], the Boltzmann equation is defined as

$$\underbrace{\frac{\partial(nf)}{\partial t}}_1 + \underbrace{\bar{C} \cdot \frac{\partial(nf)}{\partial \bar{r}}}_2 + \underbrace{\bar{a} \cdot \frac{\partial(nf)}{\partial \bar{C}}}_3 = \underbrace{\int_{-\infty}^{\infty} \int_0^{4\pi} n^2 \{f(\bar{C}')f(\bar{Z}') - f(\bar{C})f(\bar{Z})\} g\sigma d\Omega d\bar{Z}}_4 \quad . \quad (1)$$

Terms 1-3 represent transport terms of the absolute distribution function nf where n is the number density of particles at location $\bar{r}$ and time t and f is the velocity distribution function describing the probability of finding a particle in $[\bar{C}, \bar{C} + d\bar{C}]$. Term 1 builds the temporal evolution, term 2 the convection of particles in physical space and term 3 the acceleration of particles in velocity space due to an external force e.g. an electromagnetic force. The right side of the equation i.e. term 4 is also referred to as collision operator and can further be divided into the gain term $f(\bar{C}')f(\bar{Z}')$ and the loss term $f(\bar{C})f(\bar{Z})$ representing the probability of collisions creating and, respectively, destroying particles with velocity $\bar{C}$ scaled by the collision frequency $n^2 g\sigma$. Here, the apostrophe marks the post-collisional particle velocity and $\bar{Z}$ the velocity of the collision partner. The collision operator considers the particles' collision cross section σ and angle Ω . As the equation has an integral-differential structure and a large number of independent variables, finding a solution for a representative number of particles for engineering applications is incredibly computationally expensive.

Maxwell derived the equation of change [2] from the Boltzmann equation. The equation of change builds up to the macroscopic fluid descriptions like Euler or Navier-Stokes equations. According to Boyd [2],

$$\frac{\partial}{\partial t}(n\langle Q \rangle) + \frac{\partial}{\partial \bar{r}}(n\langle \bar{C}Q \rangle) = [\dot{Q}] \quad (2)$$

with $Q(\bar{C})$ as a particle property and $\dot{Q}$ as the rate of change of Q due to collisions. To connect the molecular picture with macroscopic fluid properties, moments are calculated from the Boltzmann equation. Moments can be considered as weighted averages of the distribution function [3] which means the velocity distribution function f is multiplied with v^n and then integrated over the velocity space to calculate the n -th moment. The conservation equations are derived from the first three moments as this allows to set $\dot{Q} = 0$ as shown by Bird [4]. The second term on the left side of the equation of change introduces a dependency of Q on the next higher moment $\langle \bar{C}Q \rangle$ which creates a closure problem if $\dot{Q} \neq 0$ [2], which is the case for all moments of order three and larger.

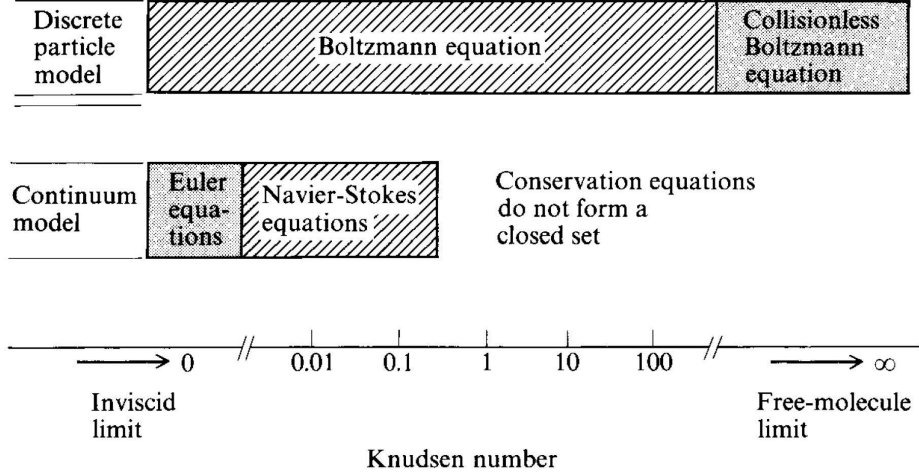


Figure 1: The Knudsen number limits on the fluid models, taken from [5] and originally from [4].

Closure can be guaranteed by imposing assumptions on the fluid's equilibrium state. The Chapman-Enskog expansion addresses this closure problem by approximating the velocity distribution function as a small perturbation from the Maxwellian. The Maxwellian velocity distribution function forms if local thermodynamic equilibrium is assumed. This implies immediate equilibration which is observed for $Kn \rightarrow 0$. The Knudsen number Kn is a measure for the degree of rarefaction and is applied to check the validity of the approximation for the Boltzmann equation.

$$Kn \equiv \lambda/L \quad (3)$$

with the mean free path length λ to the characteristic length L of a body exposed to the flow. The Chapman-Enskog method expands in the Knudsen number and allows to obtain either Euler, Navier-Stokes-Fourier, Burnett or more accurate macroscopic descriptions depending on the chosen order of approximation. According to Struchtrup [3],

$$f = f^{(0)} + \epsilon f^{(1)} + \epsilon^2 f^{(2)} + \epsilon^3 f^{(3)} + \dots \quad (4)$$

with $\epsilon \sim Kn$ and $f^{(0)}$ being the Maxwellian velocity distribution function. Struchtrup [3] presents how the Chapman-Enskog method is applied to obtain the Euler, Navier-Stokes-Fourier etc. from the Boltzmann equation. Summarizing, the Euler equations are obtained if the velocity distribution function $f = f^{(0)}$ is used and similarly the Navier-Stokes-Fourier equations are obtained for $f = f^{(0)} + \epsilon f^{(1)}$.

As the Chapman-Enskog theory implies that the velocity distribution function f is a perturbation of the Maxwellian function f_M , the perturbation from local thermal equilibrium must be sufficiently small so that f can be approximated by the chosen minimum number of moments. This small perturbation from the Maxwellian can also be expressed as a steepness limit for a fluid property gradient. A fluid property's gradient decreases with an increase in collision frequency. This is achieved with a shorter distance a particle travels between two collisions assuming a constant translational velocity. Bird [4] defines the Knudsen threshold up to which the continuum approach holds true to be $Kn = 0.1$. Boyd and other more recent literature require $Kn \lesssim 0.01$ for the continuum assumption with first order Chapman-Enskog expansion to be valid [2], [5], [6]. Methods to extend the regime of validity are further discussed in Section 2.1.2.

Figure 1 displays the full Knudsen regime with the applicability of various fluid description concepts. The following section will present for each model the governing equations including assumptions and the resulting limitations.

Boltzmann Equation Solvers The Boltzmann solver solves the Boltzmann equation (cf. Eq. 1). Due to its integral-differential shape and seven independent variables ($\mathbf{r}, \mathbf{v} \in \mathbb{R}^3$ and t) this comes with large computational costs especially for low Knudsen numbers. In general, three classes of Boltzmann solvers can be identified:

1. Full Boltzmann solvers: The collisional operator of the Boltzmann equation (cf. Eq. 1 (4)) is not approximated resulting in highest computational costs, but offering the most complete kinetic fluid description.
2. Direct Simulation Monte Carlo (DSMC): Developed by Bird [7], the continuous velocity distribution is replaced with a cloud of simulated particles, each representing many real gas molecules. This creates a statistically correct representation of the fluid. According to the algorithm's sequence, first, the particles are moved ballistically according to their velocity vector to then evaluate the impact of collisions. By averaging quantities across the discrete space, macroscopic flow variables are retrieved from the microscopic state [6], [7]. DSMC is considered the most common approach to model rarefied gases.
3. others: As the characteristics of other solver types like deterministic solvers or simplified solvers using e.g. the Bhatnagar-Gross-Krook model or the Fokker-Planck [8], [9] equation to approximate the collisional term exceed the scope of this work, no further details are provided.

2.1.2 Continuum

The connection between the microscopic particle picture drawn by the Boltzmann equation and the continuum-assuming Euler and Navier-Stokes equations is explained in Section 2.1.1. This section serves to summarize the governing equations together with the assumptions and the limitations following from Section 2.1.1.

Euler Equations The Euler equations model a fluid in local thermodynamic equilibrium. This results from the fact that the particles' velocity distribution is approximated as Maxwellian. In order for this to be a valid assumption, the number of particle collisions in the considered domain must be high to allow for a fast equilibration. Deriving from the Chapman-Eskog expansion, Boyd [2] and Anderson [5] limit the application of this assumption to $Kn \leq 0.001$ (cf. Figure 1). The consequence of the velocity distribution approximated as Maxwellian is no viscous stresses and no diffusive heat flux as both microscopic velocity and thermal gradients cannot exist. A further conclusion is that Euler equations are not capable of modeling boundary layer effects. The governing equations are defined as

$$\text{mass conservation:} \quad \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (5)$$

$$\text{momentum conservation:} \quad \frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) + \nabla p = 0 \quad (6)$$

$$\text{energy conservation:} \quad \frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{u}] = 0 \quad (7)$$

$$\text{ideal gas law:} \quad p = \rho RT \quad (8)$$

with ρ as the fluid density, $\mathbf{u}$ as the velocity vector, p as the isotropic pressure, $E = \rho e + \frac{1}{2}\rho|\mathbf{u}|^2$ as the total energy density and e is the internal energy per unit mass.

Navier Stokes Equations The Navier-Stokes equations are derived from the Boltzmann equation with first order Chapman-Enskog expansion model fluids near local thermodynamic equilibrium (cf. Section 2.1.1). Small perturbations from the Maxwellian are allowed as the

resulting effects are approximated by first order gradients to account for heat flux and viscosity. This form of the Navier-Stokes equations is also known as Navier-Stokes-Fourier as Fourier's Law results from the approximation. Anderson [5] and Boyd [2] agree the continuum assumption to be valid for the Navier-Stokes equation for $Kn < 0.01$. The governing equations are defined as

$$\begin{aligned}
\text{mass conservation:} \quad & \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \\
\text{momentum conservation:} \quad & \frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = \nabla \cdot \boldsymbol{\tau} \\
\text{energy conservation:} \quad & \frac{\partial E}{\partial t} + \nabla \cdot [(E + p)\mathbf{u}] = \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{u}) - \nabla \cdot \mathbf{q}
\end{aligned} \tag{9}$$

with

$$\begin{aligned}
\text{viscous stress tensor:} \quad & \boldsymbol{\tau} = \mu \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right] \\
\text{total energy per unit volume:} \quad & E = \rho e + \frac{1}{2}\rho |\mathbf{u}|^2 \\
\text{Fourier's law of heat conduction:} \quad & \mathbf{q} = -k \nabla T \\
\text{ideal gas law:} \quad & p = \rho R T
\end{aligned} \tag{10}$$

and μ as the viscosity as well as k as the thermal conduction coefficients.

In order to extend the validity to higher Knudsen numbers, boundary conditions must consider slip effects [5], [6]. Rarefaction effects lead to particles not fully thermalizing at a wall. This means neither velocity nor temperature of particles colliding with the walls will be the same as of the wall after collision. These jumps occur due to insufficient number of particle-collisions with the wall [2]. Under the consideration of these slip effects, the Navier-Stokes equations remain valid up to $Kn \leq 0.2$ [5] (cf. Figure 1). Boyd [2] refers to the Maxwell slip velocity boundary condition as

$$u_t = \frac{2 - \alpha_M}{\alpha_M} \frac{2\lambda_\omega}{3} \frac{\partial u}{\partial r_n} \tag{11}$$

with u_t as the particle's tangential velocity component, λ_ω as the Knudsen number in the wall regime, α_M as the tangential momentum accommodation coefficient and shows that for small $\lambda_\omega/\Delta r_n$, $u_t \rightarrow 0$ to recover the no-slip condition valid in $Kn \leq 0.01$. Boyd [2] refers to the Smoluchowsky temperature jump condition as

$$T_\omega - T_s = \frac{2 - \alpha_T}{\alpha_T} \frac{2\lambda_\omega}{3} \frac{\partial T}{\partial r_n} \tag{12}$$

with T_ω as the gas temperature at the wall, T_s as the surface wall temperature and α_T as the thermal accommodation coefficient.

If the Knudsen number increases further, the first order Chapman-Enskog expansion decreases in accuracy. Physically, thermal inequilibrium starts to develop due to the decreased number of particle-particle collisions. Thermal equilibrium refers to the particle's energy modes to not hold the same amount of energy. For the Navier-Stokes equations to remain accurate enough, a separate model for the inner energy modes needs to be added e.g. the Two-Temperature model (cf. Section 2.2.2). This generalizes Fourier's law of heat conduction so that heat flux depends on the amount of energy stored in each mode. Literature agrees that once $Kn \geq 0.2$, the continuum assumption for the Navier-Stokes equation is not sufficiently accurate and refers to either hybrid CFD-DSMC solvers (cf. Section 2.1.4) or kinetic solvers (cf. Section 2.1.1). The Navier-Stokes equations can be extended to capture specific effects by adding more accurate physics from the other orthogonal modeling dimensions as presented in Sections 2.2, 2.3 and 2.4.

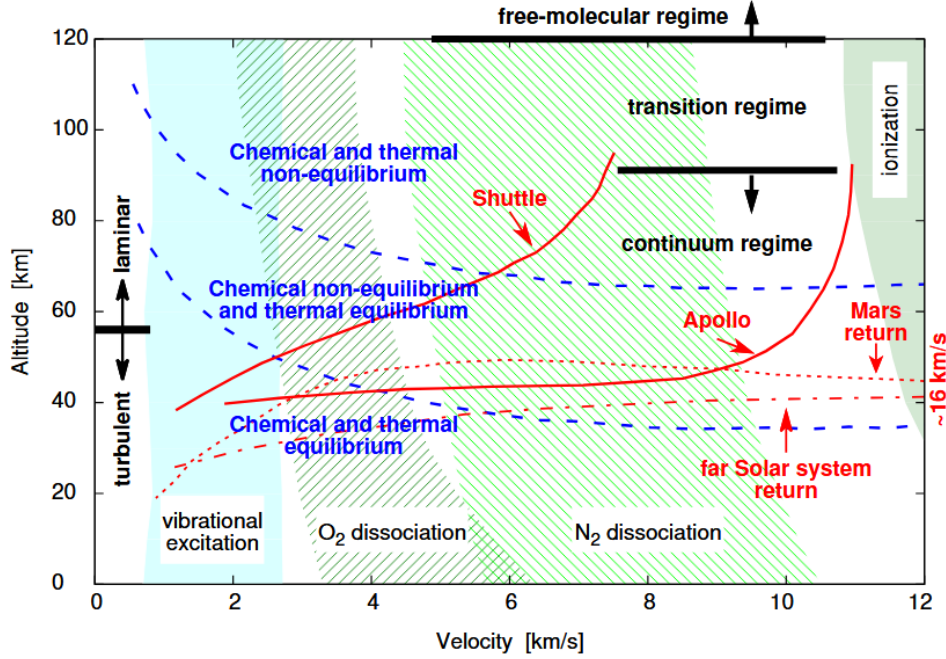


Figure 2: Variety of planetary reentry corridors with the regimes of different equilibria and reaction states visualized, taken from [6]

2.1.3 Extended Continuum

The Extended Continuum models mainly distinguish from Euler and Navier-Stokes equation from their velocity distribution function which approximates with higher order expansion terms according to the Chapman-Enskog method (cf. Section 1 and Equation 4). The so-called Burnett equations result from inserting the approximation

$$f = f^{(0)} + \epsilon f^{(1)} + \epsilon^2 f^{(2)} \quad (13)$$

into the Boltzmann equation. This promises a more accurate approximation of viscous stresses and heat flux due to higher order derivatives extending the validity of the continuum assumption in the Knudsen regime. Though, these equations are not widely used as the higher fidelity comes with the cost of largely increased instability [5].

2.1.4 Hybrid CFD-DSMC

During planetary reentry scenarios, a vehicle passes through regimes of strong chemical or thermal non-equilibria at densities which are high enough to result in prohibitive computational effort if simulated with DSMC¹ (cf. Figure 2). However, the strong non-equilibria would, on the other hand, result in large imprecisions as the assumption of small local thermodynamic equilibrium imposed by the Chapman-Enskog expansion is not valid anymore. This increases interest in hybrid solutions which break down the spatial domain into continuum and non-continuum areas to be then solved with CFD and DSMC approaches respectively. Casseau discusses the development of *hy2Foam* as such a hybrid solver in [6] with the motivation to apply it to reentry scenarios in this transition zone in the Knudsen regime.

2.2 Internal Energy Modeling

The total energy of a fluid is not only stored in the molecules' kinetic energy, but also in their translational, rotational, vibrational, electronic and chemical energy modes. According to

¹Casseau [6] reports a scaling of DSMC's computational costs proportional to Kn^{-4} .

Casseau [6], the total energy equals

$$E = \frac{1}{2}\rho \sum_i u_i^2 + \sum_{s \neq e} E_{t,s} + \sum_{s \neq e} E_{r,s} + \sum_{s \neq e} E_{v,s} + \sum_{s \neq e} E_{el,s} + E_e + \sum_{s \neq e} \rho_s h_s^\circ \quad (14)$$

composed of the kinetic energy in all three spatial dimensions i , translational, rotational, vibrational, electronic, electron and chemical energies. Here, $\rho_s h_s^\circ$ is the standard enthalpy for formation of species s . At low temperatures, the total energy is distributed equally among all energy modes also referred to as thermal equilibrium. Energy equilibration occurs via particle collision. The quantum mechanical explanation is presented by Anderson [5]. The larger the Knudsen number, the faster the internal energy modes deviate from equilibrium. At high temperatures, non-equilibrium between these modes becomes relevant to accurately predict aerodynamic coefficients as well as convective and radiative heating [10]. More detailed models are required as this non-equilibrium impacts dissociation and recombination as well as ionization processes at very high temperatures.

As the equilibration depends on the collision frequency, this process occurs at finite rate. The Damköhler number as defined by Stemmer [11] informs about how fast the equilibration of a specific energy mode occurs compared to the flow time scales:

$$Da = \frac{t_f}{\tau} \quad (15)$$

with the time t_f , a particle needs to cross a shock wave, for example, and the time τ required for the particles' energy modes to equilibrate. If $Da \gg 1$, equilibration occurs instantly compared to the flow time scales and considering the finite rate of equilibration is negligible. This also means, the energy of the flow can be modeled by a single temperature variable. For $Da \ll 1$ the flow is considered frozen. Flow conditions change so quickly that energy modes do not equilibrate. This state requires a representation of the energy with different temperature variables. $Da \approx 1$ indicates that the equilibration process and the flow convection occur at similar time scales. The finite rate of this process must be modeled for the most accurate result. The following sections provide an overview about different temperature models. The focus lies on temperature models applicable to macroscopic flow descriptions like the Navier-Stokes-Fourier equations. For the sake of completeness, section ?? lists a selection of internal energy models for kinetic models.

2.2.1 Single Temperature Model

A Single-Temperature model is applicable to flow problems with $Da \gg 1$. It assumes the instantaneous equilibration of all energy modes. These conditions occur usually in continuum flows with sufficiently low Mach and moderate temperatures. The model fails when translational-vibrational relaxation is slow compared to flow time scales. This needs to be expected for planetary re-entry scenarios (cf. Fig. 2).

2.2.2 Two-Temperature Model

The Two-Temperature model assumes that all energy modes of all reentry-relevant species (O, N, O₂, N₂, NO, N⁺, O⁺, O₂⁺, N₂⁺, NO⁺ and electrons) can be partitioned into two temperatures with acceptable loss of accuracy [10]. This assumes vibrational-electronic energy modes and the translational energy of free electrons to equilibrate instantly. Then, one temperature T describes the energy stored in the translational and rotational energy modes of all particles, and another temperature T_V approximates the vibrational, electronic and electron energies [12].

A justification with explanation at quantum state level is given by Anderson [5]. To capture the finite relaxation between modes source terms are derived from kinetic theory as shown by Gnoffo [10]. Introducing this second temperature variable T_V allows appropriate flow modeling at high-speed in transitional Kn regimes where translational-vibrational non-equilibrium is significant. More accurate estimations of heat flux and macroscopic flow characteristics as well as better shock structure predictions can be expected [6].

Gnoffo presents the translational and vibrational energy equation as²

$$\begin{aligned}
\underbrace{\frac{\partial}{\partial t}(\rho e_V)}_1 + \underbrace{\frac{\partial}{\partial x^j}(\rho e_V u^j)}_2 = & \underbrace{-p_e \frac{\partial u^j}{\partial x^j}}_3 + \underbrace{\frac{\partial}{\partial x^j} \left[(\eta_V + \eta_e) \frac{\partial T_V}{\partial x^j} \right]}_4 + \underbrace{\frac{\partial}{\partial x^j} \left(\rho \sum_{s=1}^{11} h_{V,s} D_s \frac{\partial y_s}{\partial x^j} \right)}_5 \\
& + \underbrace{\sum_{s=\text{mol.}} \rho_s \frac{(e_{v,s}^* - e_{v,s})}{\langle \tau_s \rangle}}_6 + \underbrace{2\rho_e \frac{3}{2} \bar{R} (T - T_V) \sum_{s=1}^{10} \frac{\nu_{es}}{M_s}}_7 - \underbrace{\sum_{s=6}^{10} \dot{n}_{e,s} \hat{I}_s}_8 \\
& + \underbrace{\sum_{s=\text{mol.}} \dot{w}_s \hat{D}_s}_9 - \underbrace{Q_{\text{rad}}}_{10} .
\end{aligned} \tag{16}$$

The following subsections present how the finite rate relaxation between the two temperatures can be modeled and give context to terms 6 and 7 of Equation 16 (cf. Sections 2.2.2 and 2.2.2). Term 8 is explained in Section 2.3.4, term 9 in Section 2.3.3. Radiation is addressed in Section 2.4.1.

Vibrational-Translation Relaxation Term 6 in Equation 16 is also referred to as Landau-Teller equation [6] and includes the translational-vibrational relaxation time $\langle \tau_s \rangle$ for molecule species s . Gnoffo calculates this from²

$$\langle \tau_s \rangle = \tau_s^{MW} + \tau_s^P \tag{17}$$

with τ_s^{MW} obtained from Millikan's and White's semi-empirical correlations² which are valid up to 8000K

$$r_s^{MW} = \frac{\frac{1}{p} \sum_{j=1}^{10} n_j \exp \left[A_s \left(T^{-1/3} - 0.015 \mu_{s_j}^{1/4} \right) - 18.42 \right]}{\sum_{j=1}^{10} n_j} \tag{18}$$

and τ_s^P from Park for temperatures above 8000K defined as²

$$\tau_s^P = (\sigma_s \bar{c}_s n_s)^{-1} . \tag{19}$$

Vibrational-Vibrational Relaxation The work presented by Gnoffo [10] does not mention the vibrational-vibrational relaxation process, but Casseau [6] takes it into account (cf. [6] Equation 2.65) noting this process to have a secondary role in the vibrational energy exchange.

²Explaining the meaning of all symbols is omitted from this text to maximize compactness and not distract the reader from the key message. The symbols including their indices were not changed from Gnoffo's original notation in [10].

Electron-related Relaxation Term 7 in Equation 16 uses the effective collision frequency ν_{es} of a free electron with another species s and is calculated differently for collisions with ions and neutrals (cf. [10] Equation 64-66). Other references like Casseau [6] also mention energy transfers between free electrons and the vibrational energy mode and the removal of vibrational-electronic energy due to electron impact ionization, but does not consider it as an energy source term in their work.

Chemistry-Vibrational Coupling Term 9 in Equation 16 represents the amount of vibrational energy created or destroyed at dissociation or recombination and will be explained in more detail in Section 2.3.4.

Other Extensions The Two-Temperature model increases fidelity of models for the transitional regime without introducing large complexity or computational costs. Though, there are various extensions possible which become mostly necessary if the thermal non-equilibrium grows stronger. Casseau promotes an individual vibrational energy pool for each major species (O_2 , N_2 , NO) to model each species' individual vibrational-electronic temperature where their ion counterparts are assumed to have the same temperature as the respective neutrals [6], although they note that "in most cases" the different energy pools will be approximately equal. Park [12] adds that the accuracy of a multivibrational model is strongly dependent on the availability for relaxation time data.

At high altitudes with high reentry velocities where temperatures exceed 10000K[5], molecule ionization gains in relevancy (cf. Fig. 2). For this application, Multi-Temperature models might be preferred which offer even higher internal energy resolution. A Three-Temperature model considers the free-electron temperature as an independent variable. This additional cost is traded with higher complexity.

2.2.3 State-to-State Model

A State-to-State model offers even higher resolution of the internal energy's distribution by tracking the population of all quantum states for every species' energy mode as an individual pseudo-species with their own continuity equation and source terms. While this approach models with highest accuracy, the computational costs to solve this model for a representative problem is unrealistic as explained by Maninder [13].

2.3 Chemical Kinetics Model

Chemical kinetics allows to model the evolution of species composition in fluids. Depending on the flow state i.e. degree of non-equilibrium, Damköhler number and temperature among others, different models must be used to achieve the optimum in required level of fidelity and compute expense. The following sections give an overview about different models.

2.3.1 Equilibrium Chemistry

Equilibrium chemistry assumes that chemical reactions occur infinitely fast. The species which the fluid is composed of, are lumped so that no species transport equations are needed. This imposes limitations to this model as at transitional or high Knudsen numbers, the equilibrium grows too strong to not be represented by the model on relevant timescales. In general, the model can be applied for $Da \gg 1$ (cf. Eq. 15).

2.3.2 Frozen Chemistry

For frozen chemistry models, the assumption is made of infinitely slow reaction rates. This allows all source terms in the energy equations ($\dot{\omega}_s$ in Eq. 16) to be neglected. Thus, only the

transport of species, but not evolution of their population is captured. This assumption is only valid for $Da \ll 1$.

2.3.3 Finite Rate Chemistry

Finite rate chemistry models capture flows for which $Da \approx 1$ is true. The simpler models usually represent 5 air species (O, N, O₂, N₂, NO), but do not account for ionization [14]. Finite rate effects impact the governing equations via an energy source term $\sum_{s=mol} \dot{\omega}_s \hat{D}_s$ (cf. Eq. 16 term 9) where for each species s the mass rate of production $\dot{\omega}_s$ is multiplied with the vibrational energy per unit mass $\hat{D}_s$ created or destroyed during dissociation. The mass rate of production is defined by Gnoffo [10] as³

$$\begin{aligned}\dot{\omega}_s &= M_s \sum_{r=1}^{N_r} (\beta_{s,r} - \alpha_{s,r}) (R_{f,r} - R_{b,r}), \\ R_{f,r} &= k_{f,r} \prod_{s=1}^{11} \left(\frac{\rho_s}{M_s} \right)^{\alpha_{s,r}}, \\ R_{b,r} &= k_{b,r} \prod_{s=1}^{11} \left(\frac{\rho_s}{M_s} \right)^{\beta_{s,r}}\end{aligned}\tag{20}$$

where the calculation of the reaction rate coefficients $k_{f,r}$ and $k_{b,r}$ is well outlined by Casseau including a clear overview about which rate controlling temperature to use for which reaction type (cf. [6], Tab. 2.1). In short, these reaction rate coefficients are determined based on the Arrhenius equation using parameters obtained either from experimental studies [10], or from DSMC simulations as shown by Bird [15] and Casseau [6].

To determine the vibrational energy per unit mass $\hat{D}_s$ created or destroyed during a dissociation, the literature proposes two different theories about the coupling mechanism between chemistry and vibrational energy. Preferential dissociation suggests that molecules in higher vibrational states dissociate more easily, therefore, dissociation events preferentially remove high- T_V molecules. The two most prominent models used are 1. Park's TTV model and 2. the coupled vibration-dissociation-vibration model [6], [10]. In the scope of this document, only Park's model shall be briefly touched upon. In short, Park [16] presents dissociation temperature which represents a local average of the temperature variables as

$$T_q = T^{0.7} T_V^{0.3}\tag{21}$$

which Casseau [6] only uses to calculate the reaction rate coefficient for a dissociative reaction in forward direction. To calculate the vibrational energy per unit mass $\hat{D}_s$ created or destroyed during a dissociation, Park proposes

$$\hat{D}_s = \hat{c} D_m\tag{22}$$

where D_m refers to the dissociative potential and $\hat{c}$ to a scaling factor. Park suggests a constant value of $\hat{c} = 0.3$ [10], though the more modern work by Casseau points to a linear correlation between $\hat{c}$ and T [6].

Lastly, non-preferential dissociation assumes the dissociation probability to be independent of the vibrational state, but instead simply assumes dissociation to occur at the average vibrational energy. According to Casseau [6]

$$\hat{D}_s = e_{v,m} \quad .\tag{23}$$

A clear summary including equations required to calculate various coefficients is presented in [17].

³Explaining the meaning of all symbols is omitted from this text to maximize compactness and not distract the reader from the key message. The symbols including their indices were not changed from Gnoffo's original notation in [10].

In recent work by Bird [15], a quantum-kinetic model (cf. Sec. 2.3.6) is used to compute Arrhenius parameters and argues that the preferential dissociation emerges from the kinetic model and is, thus, proven to be the right approach.

2.3.4 Multi-Species Ionization & Recombination

In typical earth reentry scenarios, the post-collisional transitional temperatures of heavy particles are usually not exceeding the ionization energies of air molecules [6]. Though, ions are also created by other mechanisms. Associative ionization where two neutrals collide to form an ion and an electron ($A + B \rightarrow AB^+ + e^-$), requires far less energy because the association releases binding energy facilitating an electron release. However, Gnoffo [10] only models Electron-impact ionization to account for energy loss when free electrons hit other neutrals leading to ionization and a reduction of electron energy (cf. Eq. 16, term 8) and even here focuses only on electron impact ionization of O and N. Via backward reactions these ions will then be consumed to recombine into other ion species like O_2^+ , N_2^+ or NO^+ . In total, finite-rate chemistry models with ionization for reentry scenarios in their most common form model 11 species (O, N, N_2 , O_2 , NO, O^+ , N^+ , N_2^+ , O_2^+ , NO^+ and e^-).

2.3.5 State resolved chemistry

In state-resolved chemistry, each internal quantum level of a molecule is treated as a separate species (cf. Section 2.2.3). This results in a large number of state variables and is recommended only for large non-equilibria.

2.3.6 Quantum-Kinetic model

The key difference between the Quantum-Kinetic model and the continuum-based chemistry model presented in Section 2.3.3 is that it models the dependency of chemical reactions on the actual collision energy and actual quantum vibrational level, and not on macroscopic temperatures [15]. Interestingly, preferential dissociation emerges from the model and is not a model assumption as discussed for the finite rate chemistry models in Section 2.3.3. Another major advantage the Quantum-Kinetic model offers is its independence on empirical Arrhenius parameters which are required for Park’s TTV model (cf. [10], Eq. 46a) [6], [15]. This allows in return to compute potentially more accurate Arrhenius parameters from the quantum-kinetic model than are currently available from experimental studies, as shown by Casseau [6].

2.4 Additional Physical Subsystems

Additional physics can be considered as required. Due to highly elevated temperatures, the electronic states might be excited highly enough that radiation plays a considerable role (cf. Section 2.4.1). Anderson sets this limit at 10000K⁴ [5]. Another effect occurring only significant at high enough temperature is ionization. With a large enough degree of ionization, plasmadynamic effects need to be considered in order to maintain high enough accuracy. For N_2 and O_2 ionization sets in at 9000K [5]. Depending of the objective of the study, more sophisticated wall chemistry like ablation could be considered. The following briefly touches on radiation.

2.4.1 Radiation

Radiation in the form of photons is created by electrons changing their excitation state. In general, this can happen through four different transition types: bound-bound, bound-free, free-bound and free-free transition. The naming refers to the electron state before and after

⁴Orbital reentry maneuvers usually do not exceed reentry velocities of 10 km/s. This velocity spectrum starts to become relevant for returns from lunar, martian or even deeper missions (cf. Fig. 2).

the transition, where "bound" means the electron to be bound to an atom and "free" that the electron exists unbounded. Each transition type results in a different radiation spectrum. Modeling radiation implies high computational costs as the fluid needs to be modeled with state-to-state accuracy (cf. Section 2.2.3). Even for a small number of atoms modeled a large spectral fidelity is required in order to capture narrow spectral lines to be able to compute the radiated flux correctly. To model radiation at highest fidelity, the result of the radiation solver would need to feed back into the solution of the continuum solver as radiation implies a change in the electronic energy. This requires another convergence loop per time step. A simplification can be done, but just computing the radiation as a post-processing step for every time step. Casseau [6] neglects radiation effects in his work and suggests as a future work to couple with radiation solvers. Yang [18] develops a simplified, uncoupled model to compute radiated power, though, also limits the scope to flight velocities above 10 km/s.

3 Governing Equations

This section presents the physics implemented to compute the results presented in the following Section 5. Independent of Section 2, all relevant equations are brought into one coherent context. These are the conservation equations, the different models for internal energy, the computation of the transport coefficients and source terms.

3.1 Conservation Equations

The Navier-Stokes-Fourier-Equations according to Equation 9 are extended to model thermal and chemical non-equilibrium. This extensions allows more accurate modeling of hypersonic effects [6], [10], [19]. The extension includes the consideration of two temperatures. T expresses the excitation of the translational and rotational energy mode while T_V describes the energy stored in the vibrational, electronic and electron translational energies of all species representing the fluid. This split is useful to model the difference in quantization of these energy modes. This affects how quickly energy equilibrates across the different modes once excited by an external force like a particle collision. At time scales of hypersonic physics, this equilibration can generally not be considered as instant. Though, it has an effect on dissociation of polyatomic molecules [6], [10]. The model for the thermal equilibration process is presented in Section 3.3. Another extension to the model is the consideration of the finite rate of chemical processes. Hypersonic physics occur at time scales on which chemical equilibrium is generally not reached instantly on flow time scales. This behavior is modeled with Arrhenius rates to compute the source term contribution for the mass and vibroelectronic energy conservation equations and is presented in Section 3.4.

The following conservation equations are adopted from Gnoffo [10] and Casseau [6] omitting ionization effects. The considered species are O, N, O₂, N₂ and NO. All source term contributions including charged particles are not considered as part of this work. Also, electronic energy losses due to radiation will be ignored to constrain the scope of this work to realistic magnitudes. This means scenarios with extreme fluid temperatures would not be accurately modeled by the set of equations. For air, this limit is about 10000K [5].

Following the Two-Temperature model according to Gnoffo [10], only two energy conservation equations are solved while Casseau [6] proposes one vibrational energy conservation equation per diatomic species. Thus, the total number of conservation equations equals $N = n_s + D + 2$ with n_s as the number of considered species and $D \in \{1, 2, 3\}$ as the number of spatial dimensions.

mass:

$$\frac{\partial \rho_s}{\partial t} + \nabla \cdot (\rho_s \mathbf{u}) = \nabla \cdot (\rho D_s \nabla y_s) + \dot{\omega}_s \quad (24)$$

momentum:

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u} + p \mathbf{I}) = \nabla \cdot \boldsymbol{\tau} \quad (25)$$

vibroelectronic energy:

$$\begin{aligned} \frac{\partial(\rho e_V)}{\partial t} + \nabla \cdot (\rho e_V \mathbf{u}) = & \nabla \cdot (\eta_V \nabla T_V) + \nabla \cdot \left(\rho \sum_s h_{V,s} D_s \nabla y_s \right) \\ & + \sum_m \rho_m \frac{e_{V,m}^* - e_{V,m}}{\langle \tau_m \rangle} + \sum_m \dot{\omega}_m \hat{D}_m \end{aligned} \quad (26)$$

total energy:

$$\begin{aligned} \frac{\partial(\rho E)}{\partial t} + \nabla \cdot [(\rho E + p)\mathbf{u}] &= \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{u}) + \nabla \cdot (\eta \nabla T + \eta_V \nabla T_V) \\ &+ \nabla \cdot \left(\rho \sum_s h_s D_s \nabla y_s \right) \end{aligned} \quad (27)$$

with

$$\text{viscous stress tensor: } \boldsymbol{\tau} = \mu \left[\nabla \mathbf{u} + (\nabla \mathbf{u})^T - \frac{2}{3}(\nabla \cdot \mathbf{u})\mathbf{I} \right] \quad (28)$$

$$\text{total energy per unit volume: } E = \sum_s \frac{\rho_s e_s}{\rho} + \frac{1}{2}|\mathbf{u}|^2 \quad (29)$$

and μ as the viscosity, η and η_V as the transrotational and vibroelectronic thermal conduction coefficients, respectively, and the subscripts s and m to indicate either all species or only all molecules i.e. no single-atom species. Adapted from Gnoffo [10], the mixture mass density ρ is calculated as

$$\rho = \sum_{s=1}^5 \rho_s \quad , \quad (30)$$

the species mole fraction y_s as

$$y_s = \frac{\rho_s/M_s}{\sum_{k=1}^5 \rho_k/M_k} \quad , \quad (31)$$

and the pressure p as

$$\begin{aligned} p &= \sum_{s=1}^5 p_s \\ p_s &= \rho_s \bar{R}T/M_s \quad . \end{aligned} \quad (32)$$

The computation of e , e_V , h and h_V in Section 3.2, of $\langle \tau_m \rangle$ in Section 3.3, of $\dot{\omega}_m$ and $\hat{D}_m$ in Section 3.4 and of μ , η , η_V and D_s is explained in Section 3.5.

3.2 Thermodynamic Relations

A molecule's total internal energy e_s per unit mass of species s is defined as

$$e_s = e_{t,s} + e_{r,s} + e_{v,s} + e_{el,s} \quad (33)$$

where the subscripts specify the translational (t), rotational (r), vibrational (v) and electronic (el) energy modes. Translational and rotational energies scale proportionally with T .

$$e_{t,s} = \frac{3}{2}R_s T \quad (34)$$

$$e_{r,s} = \begin{cases} R_s T & , \text{ species is molecular} \\ 0 & , \text{ species is monoatomic} \end{cases} \quad (35)$$

To model the vibrational energy in dependence of T_V , Bird [7] proposes a harmonic oscillator

$$e_{v,s} = \begin{cases} R_s \frac{\theta_{v,s}}{\exp\left(\frac{\theta_{v,s}}{T_v}\right) - 1} & , \text{ species is molecular} \\ 0 & , \text{ species is monoatomic} \end{cases} \quad (36)$$

with the characteristic vibrational temperature $\theta_{v,s}$. The electronic energy is determined by multiplying the characteristic electronic energy $\theta_{el,i,s}$ of each electronic state i with the probability of an electron occupying this state integrated over all states i and normalized by the electronic partition function $\sum_i g_{i,s} \exp\left(-\frac{\theta_{el,i,s}}{T_V}\right)$.

$$e_{el,s} = R_s \frac{\sum_{i \neq 0} g_{i,s} \theta_{el,i,s} \exp\left(-\frac{\theta_{el,i,s}}{T_{V,s}}\right)}{\sum_i g_{i,s} \exp\left(-\frac{\theta_{el,i,s}}{T_{V,s}}\right)} \quad (37)$$

Casseau [6] provides the data for $\theta_{v,s}$, $\theta_{el,i,s}$ and $g_{i,s}$ tabulated in Table A1 and B1, respectively, for all relevant species which allows to calculate the magnitude of the energy vibrational and electronic mode independently. Gnoffo [10] only provides curve fits for the specific enthalpy h_s from all internal energy modes, including enthalpy at formation. Based on these curve fits, a split into transrotational and vibroelectronic contribution is possible, but a state-individual resolution cannot be determined. Latter might be desired depending on the dissociation model of choice. To calculate the specific equilibrium enthalpy over all inner energy modes, Gnoffo proposes

$$h_s(\bar{T}) = R_s \left(\sum_{k=1}^5 \frac{A_k^s \bar{T}^k}{k} + A_6^s \right) \quad (38)$$

and provides the parameter values for the curve fits in Table II. To determine the non-equilibrium enthalpy, first, the vibroelectronic contribution to the specific enthalpy is determined by

$$h_{V,s}(T_V) = h_s(\bar{T} = T_V) - (C_{p,t}^s + C_{p,r}^s)(T_V - T_{\text{ref}}) - h_{s,0} \quad (39)$$

with $T_{\text{ref}} = 298.16$ K, $h_{s,0}$ in provided in Tab I and

$$C_{p,t}^s = \frac{3}{2} R_s \quad (40)$$

$$C_{p,r}^s = \begin{cases} R_s & , \text{ species is molecular} \\ 0 & , \text{ species is monoatomic} \end{cases} . \quad (41)$$

With that, the non-equilibrium specific enthalpy is calculated as

$$h_s(T, T_V) = h_{V,s}(T_V) + (C_{p,t}^s + C_{p,r}^s)(T - T_{\text{ref}}) + h_{s,0} . \quad (42)$$

Both internal energy models, Bird's harmonic oscillator with the summation over all electronic levels as presented by Casseau [6] as well as the curve fit used by Gnoffo [10] are implemented in the solver structure. A comparison of the resulting degrees of freedom estimated by both models is shown in Figure 3.

For the sake of completeness, the vibrational and electronic heat capacities according for the model proposed by Casseau [6] are defined as

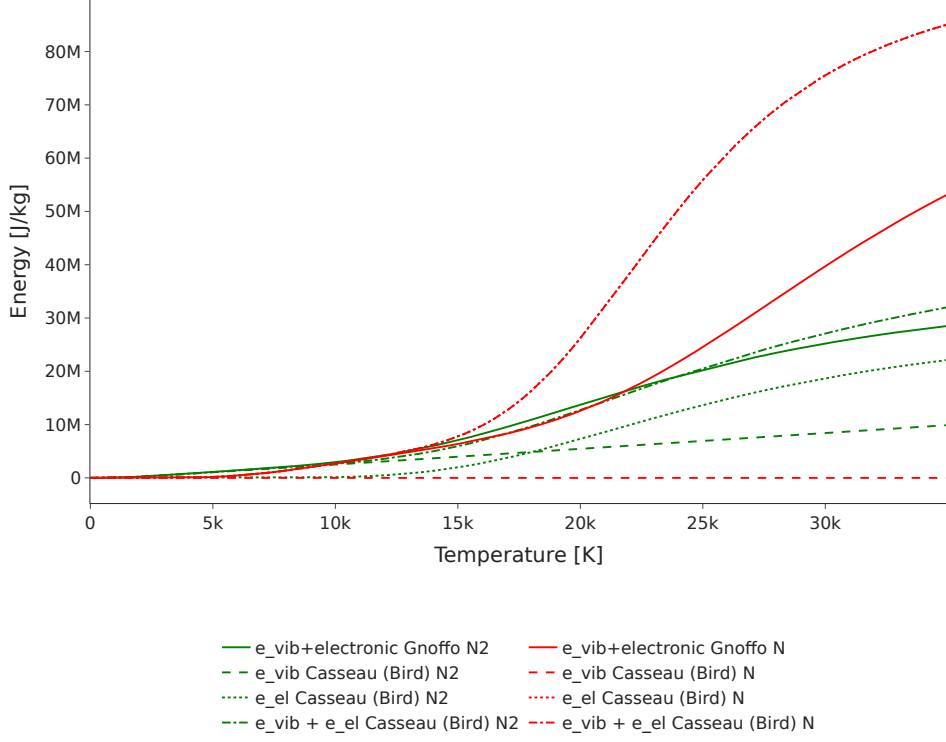


Figure 3: Comparison of Casseau's [6] and Gnoffo's [10] energy models for N_2 and N . The stark difference in e_V , especially for N , is emphasized for $T \geq 15000K$.

$$C_{v,v,s} = \frac{\partial e_{v,s}}{\partial T_V} = R_s \left(\frac{\theta_{v,s}}{T_v} \right)^2 \frac{\exp\left(\frac{\theta_{v,s}}{T_v}\right)}{\left[\exp\left(\frac{\theta_{v,s}}{T_v}\right) - 1 \right]^2} \quad (43)$$

$$C_{v,el,s} = \frac{\partial e_{el,s}}{\partial T_V} = \frac{R_s}{T_V^2} \left[\frac{\sum_i g_{i,s} \theta_{el,i,s}^2 \exp(-\theta_{el,i,s}/T_V)}{\sum_i g_{i,s} \exp(-\theta_{el,i,s}/T_V)} - \left(\frac{\sum_i g_{i,s} \theta_{el,i,s} \exp(-\theta_{el,i,s}/T_V)}{\sum_i g_{i,s} \exp(-\theta_{el,i,s}/T_V)} \right)^2 \right] \quad (44)$$

and the vibroelectronic heat capacity using Gnoffo's curve fit as

$$C_{v,V,s}(T_V) = \sum_{k=1}^5 A_{s,k} T_V^{k-1} - \begin{cases} \frac{7}{2} R_s, & \text{species is monoatomic} \\ \frac{5}{2} R_s, & \text{species is diatomic} \end{cases} \quad (45)$$

The following introduces the definition of the overall temperature T_{ov} and of the degrees of freedom $\zeta_{q,s}$ each energy mode donates. These are useful to compute while interpreting mismatches to reference data.

$$T_{ov} = \frac{\sum_s [y_s (\zeta_{t,s} + \zeta_{r,s}) T_{tr}] + \sum_s [y_s (\zeta_{v,s} + \zeta_{el,s}) T_V]}{\sum_s [y_s (\zeta_{t,s} + \zeta_{r,s} + \zeta_{v,s} + \zeta_{el,s})]} \quad (46)$$

$$\zeta_{t,s} = 3 \quad (47)$$

$$\zeta_{r,s} = \begin{cases} 2, & \text{species is molecular} \\ 0, & \text{species is monoatomic} \end{cases} \quad (48)$$

$$\zeta_{v,s} = \begin{cases} \frac{2e_{v,s}(T_V)}{R_s T_V}, & \text{species is molecular} \\ 0, & \text{species is monoatomic} \end{cases} \quad (49)$$

$$\zeta_{el,s} = \frac{2e_{el,s}(T_V)}{R_s T_V} \quad (50)$$

$$\zeta_{V,s} = \zeta_{v,s} + \zeta_{el,s} \quad (51)$$

3.3 Energy Relaxation

The vibrational-translational energy relaxation for polyatomic species m is modeled by the Landau-Teller equation defined as

$$Q_{m,V-T} = \rho_m \frac{\partial e_{V,m}(T_V)}{\partial t} = \rho_m \frac{e_{V,m}(T) - e_{V,m}(T_V)}{\tau_m} \quad (52)$$

The literature calculates the vibrational-translational relaxation time constant τ_m [6], [10], [19] based on semiempirical correlation τ_m^{MW} by Millikan and White [20] with the Park correction τ_m^P for temperatures above 8000 K.

$$\tau_m = \tau_m^{MW} + \tau_m^P \quad (53)$$

$$\tau_m^{MW} = \frac{\sum_{j=1}^5 n_j \exp \left[A_m \left(T^{-1/3} - 0.015 \mu_{mj}^{1/4} \right) - 18.42 \right]}{p \sum_{j=1}^5 n_j} \quad (54)$$

$$\tau_m^P = (\sigma_m \bar{c}_m n_m)^{-1} \quad (55)$$

where the values for A_m are tabulated in [10] TableI, p is in atm, the reduced molecular weight is $\mu_{mj} = M_m M_j / (M_m + M_j)$, the subscripts (m, j) refer to the colliding pair, the average molecular speed is $\bar{c}_m = \sqrt{8kT/\pi m_s}$, n_m is the number density of species m and σ_m is the effective cross section which Gnoffo [10] defines as

$$\sigma_m = \sigma^{Gnoffo} = 10^{-20} \text{m}^2 \quad (56)$$

while Casseau [6] proposes

$$\sigma_m = \sigma^{Casseau} \left(\frac{50000}{T} \right)^2, \sigma^{Casseau} = 10^{-21} \text{m}^2 \quad (57)$$

Scalabrin [19] uses the same structure as Casseau to calculate σ_m , but uses $\sigma^{Scalabrin} = 10^{-20} \text{m}^2$ to scale with temperature.

3.4 Chemistry

The mass rate of production or destruction $\dot{\omega}_s$ of a species s per unit volume is expressed as a sum over all reactions N_r in which the species is either a reactant or product, and is denoted as

Reaction Type	Forward, $T_{c,f}$	Backward, $T_{c,b}$
dissociation	T_P	T
exchange	T	T

Table 1: Controlling temperature depending on the reaction type

$$\dot{\omega}_s = M_s \sum_{s=1}^{N_r} (\beta_{s,r} - \alpha_{s,r})(R_{f,r} - R_{b,r}) \quad (58)$$

where $\alpha_{s,r}$ and $\beta_{s,r}$ are the stoichiometric coefficients for reactants and products in the reaction r and are documented in a reaction table e.g. by Gnoffo [10] in Table III, while $R_{f,r}$ and $R_{b,r}$ are the forward and backward reaction rates for the reaction r which are defined as

$$R_{f,r} = k_{f,r} \prod_{k=1}^5 \frac{\rho_k}{M_k}^{\alpha_{k,r}} \quad (59)$$

$$R_{b,r} = k_{b,r} \prod_{k=1}^5 \frac{\rho_k}{M_k}^{\beta_{k,r}} \quad (60)$$

The forward reaction rate constant $k_{f,r}$ is calculated based on the Arrhenius law as

$$k_{f,r}(T_{c,f}) = A_r T_{c,f}^{\beta_r} \exp\left(\frac{-T_{a,r}}{T_{c,f}}\right) \quad (61)$$

where A_r , β_r and the activation temperature $T_{a,r}$ derived from the activation energy are tabulated for each reaction by Gnoffo [10] in Table III, by Park [21] in Table 8.2 and 8.3, by Park [22] in Table 8.2 and 8.3, by Scalabrin [19] in Table A11 and A12, and by Casseau [6] derived from a quantum-kinetic model in Table C1-C15. A comparison of the for this work relevant Arrhenius parameters is found in Table 24. The different pattern sets are available to use and can be chosen individually by the user. $T_{c,f}$ is the rate controlling temperature for the forward reaction, further specified by Table 1.

The backward reaction rate constant is scaled by the equilibrium constant K_{eq} which Gnoffo defines only as a function of temperature $K_{eq} = f(T_{c,b})$ while Scalabrin also adds a number density dependence $K_{eq} = f(T_{c,b}, n)$ to the corresponding curve fits.

$$k_{f,r} = \frac{k_{f,r}(T_{c,b})}{K_{eq}(T_{c,b})} \quad (62)$$

$$K_{eq}(T_{c,b}) = \exp\left(\frac{A_1 T_{c,b}}{10^4} + A_2 + A_3 \ln\left(\frac{10^4}{T_{c,b}}\right) + \frac{A_4 10^4}{T_{c,b}} + A_5 \left(\frac{10^4}{T_{c,b}}\right)^2\right) \quad (63)$$

The different fits are provided by Gnoffo [10] in Table III and by Scalabrin [19] in Appendix A. Figure 4 shows the difference between both models.

3.4.1 Chemical-vibrational coupling

Different models have been proposed by literature to compute the rate controlling temperature for a forward dissociation reaction as well as the amount of vibroelectronic energy added or removed by a dissociation reaction. The discussion mainly focuses on whether a dissociation preferably occurs at higher vibrational temperatures, or not, i.e. is a molecule in a higher vibrational energy state more likely to dissociate. In this work, both options are available to use and compared in Section 5.2.

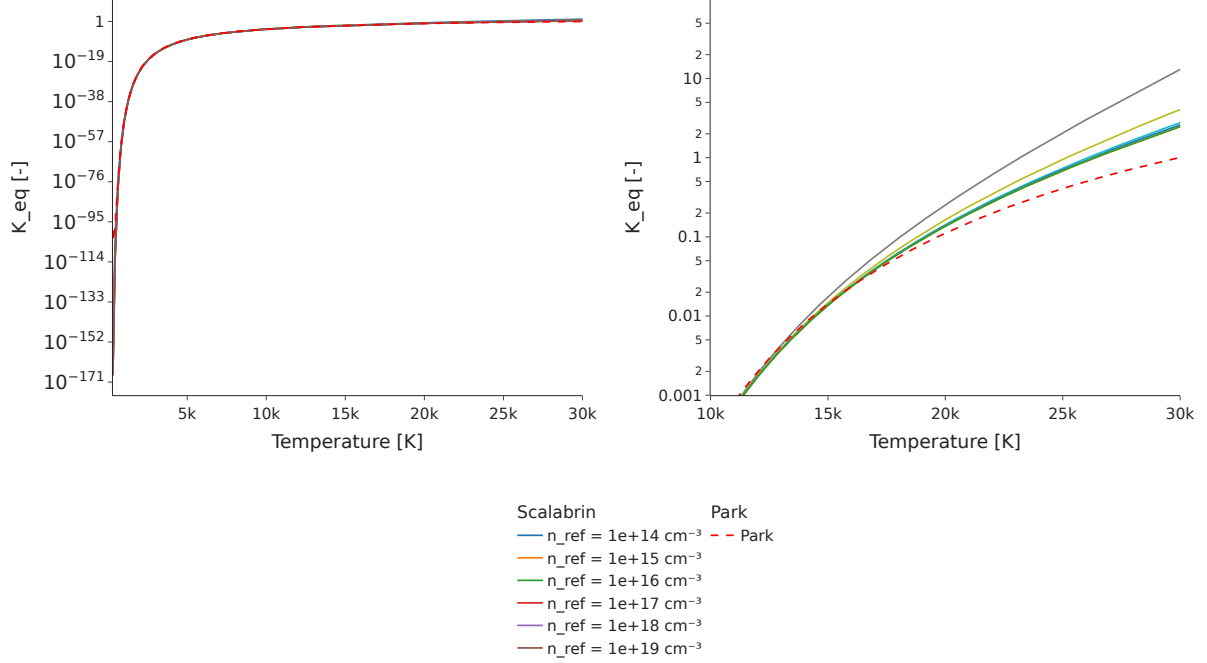


Figure 4: Comparison of Scalabrin's [19] density-dependent model (solid lines) and Park's model proposed by Gnoffo in [10] (dashed line) of the equilibrium constant K_{eq} for N_2 . *Left*: value range of K_{eq} across the full temperature range, *Right*: temperature range in which K_{eq} models diverge

Park's TTv model The rate controlling temperature according to Park's TTv model is defined as

$$T_{c,f} = T^\alpha T_V^{(1-\alpha)} \quad (64)$$

which can be understood as an average of the local temperatures where $\alpha = 0.7$ is chosen as default value according to Gnoffo [10]. The vibrational energy added or removed due to a dissociative reaction $\hat{D}_s$ per unit mass of species s is defined as

$$\hat{D}_m = D'_m + e_{el,m} \quad (65)$$

where according to the nonpreferential model

$$D'_m = e_{v,m} \quad (66)$$

and to the preferential model

$$D'_m = c_{pref} D_m \quad (67)$$

with the dissociation potential D_m tabulated by Gnoffo in Table I and by Casseau in Appendix A, and $c_{pref} = 0.3$ usually taken by literature [6].

Coupled vibration-dissociation-vibration model The coupled vibration-dissociation-vibration (CVDV) model proposed by Marrone and Treanor [23] follows a different structure than Park's TTv model. Preferential dissociation is modeled with an extension of the Arrhenius equation as

$$k_f = \frac{Z(T_{tr}) Z(T_{F,m})}{Z(T_{ve,m}) Z(-U_m)} A T_{tr}^\beta \exp\left(-\frac{T_a}{T_{tr}}\right) \quad (68)$$

with

$$U_m = \frac{D_m}{3R_m} \quad (69)$$

$$T_{F,m}^{-1} = \frac{1}{T_{ve,m}} - \frac{1}{T_{tr}} - \frac{1}{U_m} \quad (70)$$

$$Z(T) = \sum_{\alpha=0}^{\mathfrak{R}_m} \exp\left(-\frac{\varepsilon_{v,\alpha,m}}{k_B T}\right) \quad (71)$$

and α as the vibrational level, $\mathfrak{R}_m$ as the cut of value for the number of vibrational levels of species m before dissociation occurs and the energy for the α -th vibrational level as $\varepsilon_{v,\alpha,m} = \alpha k_B \theta_{v,m}$. To compute the vibrational energy source term, the model proposes

$$\bar{E}(T_{tr}, T_{ve,m}) = \frac{1}{Z(T_{F,m})} \sum_{\alpha=0}^{\mathfrak{R}_m} \varepsilon_{v,\alpha,m} \exp\left(-\frac{\varepsilon_{v,\alpha,m}}{k_B T_{F,m}}\right) \quad (72)$$

$$\bar{E}(T_{tr}, T_{tr}) = \frac{1}{Z(-U_m)} \sum_{\alpha=0}^{\mathfrak{R}_m} \varepsilon_{v,\alpha,m} \exp\left(\frac{\varepsilon_{v,\alpha,m}}{k_B U_m}\right) \quad (73)$$

$$Q_{m,C-V} = \frac{1}{m_m} [-\bar{E}(T_{tr}, T_{ve,m}) \dot{\omega}_{f,m} + \bar{E}(T_{tr}, T_{tr}) \dot{\omega}_{b,m}] \quad (74)$$

3.5 Transport

Literature proposes different approaches to determine the transport properties μ , η , η_V and D_s . The following firstly presents the Wilke's mixing rule [24], Blottner's [25] curve fits to determine the species viscosity and Eucken's [26] relation to compute the thermal conductivity. Scalabrin [19] limits the validity of this model to small speeds with flow temperature below 10000 K and small degree of ionization. Outside of this regime, Scalabrin [19] recommends to use Gupta's [27], [28] approach which is presented in Section 3.5.2. Both models have been implemented, as Casseau [6] employs the Wilke/Blottner/Eucken combination to generate the reference results used in Section 5, and Gnoffo follows Gupta's model.

3.5.1 Wilke/Blottner/Eucken

According to Wilke's [24] mixing rule

$$\mu = \sum_s \frac{y_s \mu_s}{\phi_s} \quad (75)$$

$$\eta_q = \sum_s \frac{y_s \eta_{q,s}}{\phi_s} \quad (76)$$

where μ_s is the species viscosity, $\eta_{q,s}$ the species thermal conductivity for energy mode q and ϕ_s as

$$\phi_s = \sum_r y_r \left[1 + \sqrt{\frac{\mu_s}{\mu_r}} \left(\frac{M_r}{M_s} \right)^{1/4} \right]^2 \left[\sqrt{8 \left(1 + \frac{M_s}{M_r} \right)} \right]^{-1}. \quad (77)$$

Using Blottner's [25] curve fits, the species viscosity is calculated according to

$$\mu_s = 0.1 \exp[(A_s \ln T + B_s) \ln T + C_s] \quad (78)$$

with A_s , B_s and C_s provided by Scalabrin [19] in Appendix A and by Casseau [6] in Appendix A. Using Eucken's relation, the thermal conductivities are determined with

$$\eta_{tr,s} = \frac{5}{2}\mu_s C_{v,t,s} + \mu_s C_{v,r,s} \quad (79)$$

$$\eta_{V,s} = \mu_s C_{v,V,s} \quad . \quad (80)$$

The binary diffusion coefficient D_s is calculated using the Lewis model

$$D_s = \frac{Le \times \eta}{\rho C_p} \quad (81)$$

with the Lewis number $Le = const. = \frac{\alpha}{D_s}$, η as the mixture thermal conductivity and C_p as the mixture heat capacity at constant pressure.

3.5.2 Gupta

Gupta's approach builds on the collision integrals $\pi\bar{\Omega}_{s,r}^{1,1}$ and $\pi\bar{\Omega}_{s,r}^{2,2}$ from which the mixture transport properties are derived. Based on the modified collision integrals

$$\Delta_{sr}^{(1)}(T) = \frac{8}{3} \left[\frac{2M_s M_r}{\pi R T (M_s + M_r)} \right]^{1/2} \pi\bar{\Omega}_{sr}^{(1,1)} \quad (82)$$

$$\Delta_{sr}^{(2)}(T) = \frac{16}{5} \left[\frac{2M_s M_r}{\pi R T (M_s + M_r)} \right]^{1/2} \pi\bar{\Omega}_{sr}^{(2,2)} \quad (83)$$

the mixture viscosity is calculated as

$$\mu = \sum_{s=1}^{10} \frac{m_s y_s}{\sum_{r=1}^{10} y_r \Delta_{sr}^{(2)}(T)} \quad . \quad (84)$$

The translational and rotational thermal conductivities are determined by

$$\eta_t = \frac{15}{4} k \sum_{s=1}^{10} \frac{y_s}{\sum_{r=1}^{10} a_{sr} y_r \Delta_{sr}^{(2)}(T)} \quad (85)$$

$$a_{sr} = 1 + \frac{[1 - (m_s/m_r)] [0.45 - 2.54(m_s/r)]}{[1 + (m_s/m_r)]^2} \quad (86)$$

$$\eta_r = k \sum_{s=\text{mol}} \frac{y_s}{\sum_{r=1}^{10} 10 y_r \Delta_{sr}^{(1)}(T)} \quad . \quad (87)$$

The mixture thermal conductivity for translational and rotational energy mode is determined by

$$\eta = \eta_t + \eta_r \quad (88)$$

while the vibrational thermal conductivity η_v equals η_r

$$\eta_v = \eta_r \quad . \quad (89)$$

Lastly, the effective diffusion coefficient D_s is expressed as

$$D_s = \frac{M_s (1 - M_s y_s)}{5 \sum_{\substack{r=1 \\ r \neq s}} (y_r / s_r)} \quad (90)$$

$$D_{sr} = \frac{kT}{p \Delta_{sr}^{(1)}(T)} \quad (91)$$

Note that Equations 84,87 and 91 are simplified from Gnoffo [10] and Scalabrin [19] due to the fact that the free electron mole fraction y_e will always be zero in the scope of this work as ionization is ignored.

4 Implementation

This section presents an overview about the code architecture (Section 4.1), the numerical schemes (Section 4.2) and the boundary conditions (Section 4.3) which have been implemented.

The code developed for this project can be found in the repository under this link:
https://github.com/henrihoechter/jaxfluids_internship.

4.1 Code Architecture

This section aims to give an overview about the architecture of the repository developed for this project and which different models and parameter sets are available at this point. Cases can be simulated in OD i.e. single cell transient, 1D e.g. a shock tube, or 2D where the dimensionality refers to the domain. All dimensions are calculated with the same solver. Tables 2-6 present an overview about the configuration options of the solver.

Thermodynamics		EnergyModelConfig()	
Category		Options	Notes
Internal Model	Energy	"bird", "gnoffo"	Bird: harmonic oscillator + electronic levels (NASA TP-2867); Gnoffo: polynomial fits for vibro-electronic energy
Electronic Energy		include_electronic=True/False	Toggle inclusion of electronic energy modes. Always true True for Gnoffo's model.

Table 2: Thermodynamic model configuration.

Transport		TransportModelConfig()	
Category		Options	Notes
Transport Model		"gnoffo", "casseau"	Gnoffo: Chapman–Enskog approach with collision integrals (TP-2867); Casseau: Blottner curve fits for viscosity, Eucken-type conductivity
Diffusion		include_diffusion=True/False	Enable/disable species diffusion
Collision data	integral	collision_integrals_path	Path to JSON file with species-pair collision integrals; required for "gnoffo"
Casseau data	transport	casseau_data_path	Path to JSON file with Blottner coefficients and diffusion diameters; required for "casseau"

Table 3: Transport model configuration.

Numerics	NumericsConfig()	
Category	Options	Notes
Flux scheme (1D)	"lax_friedrichs", "hllc", "exact_riemann"	–
Flux scheme (2D)	"hllc"	Only validated option
Time integration	"forward-euler", "rk2"	–
Spatial scheme	"first_order", "muscl"	
Slope limiter	"minmod", "mc"	–

Table 4: Numerical scheme configuration.

Chemistry	ChemistryModelConfig()	
Category	Options	Notes
Rate model	"park", "cvdv_qp"	Park: two-temperature Arrhenius rate; CVDV-QP: coupled vibration-dissociation model
Vibrational source	"nonpreferential", "preferential_constant"	Park model only; controls how vibrational energy is removed upon dissociation
QP constant	qp_constant (default 0.3)	Quasi-preferential constant for the CVDV-QP model
Park α	park_alpha (default 0.7)	Blending exponent for the geometric-mean temperature $T_a = T^\alpha T_v^{1-\alpha}$
Reaction data	external JSON file	Arrhenius coefficients (A , β , E_f/k), stoichiometry, and equilibrium coefficients

Table 5: Chemistry model configuration.

Boundary Conditions	BoundaryConditionArrays()	
Type	Keyword	Notes
Outflow	"outflow"	Zero-gradient ghost state.
Inflow	"inflow"	Prescribed primitive state from boundary data.
Wall	"wall"	No-slip wall with optional wall translational/vibrational temperature and optional wall composition.
Slip Wall	"wall_slip"	Maxwell slip / temperature-jump wall model.
Euler Wall	"wall_euler"	Inviscid slip / symmetry wall; also used for the axis in axisymmetric cases.
Reflective	"reflective"	Conserved-space reflection of normal momentum; mainly used for reflective 1D boundaries.
Periodic	n.a.	Periodicity is imposed via mesh connectivity.

Table 6: Boundary-condition types.

4.2 Numerics

This section touches on the solver used to compute the convective flux (cf. Section 3) as well as the important numerical schemes for spatial reconstruction, integration in time and the different boundary condition types available.

4.2.1 Harten-Lax-van Leer-Contact Solver

The Harten-Lax-van Leer-Contact (HLLC) assumes a three-wave structure of two acoustic waves and one contact discontinuity to solve a Riemann problem. With modifications presented in Section 3, the HLLC can be used to solve hypersonic flow scenarios [6]. The following equations are a condensed version of [29] and, for clarity, are presented for the one-dimensional Euler equations. To calculate the flux at cell boundaries, the following piecewise function is defined

$$\mathbf{F}_{i+\frac{1}{2}}^{\text{hllc}} = \begin{cases} \mathbf{F}_L, & \text{if } 0 \leq S_L, \\ \mathbf{F}_{*L}, & \text{if } S_L \leq 0 \leq S_*, \\ \mathbf{F}_{*R}, & \text{if } S_* \leq 0 \leq S_R, \\ \mathbf{F}_R, & \text{if } 0 \geq S_R \end{cases} \quad (92)$$

where S_L and S_R represent estimates of the left and right acoustic wave speeds, u_L and u_R as the velocities left and right, normal to the interface and S_* denotes the speed of the contact discontinuity:

$$S_L = \min(u_L, u_R) - \max(a_L, a_R), \quad (93)$$

$$S_R = \max(u_L, u_R) + \max(a_L, a_R), \quad (94)$$

$$S_* = \frac{p_R - p_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)}. \quad (95)$$

The fluxes will be calculated according to

$$\mathbf{F}_L = \mathbf{F}(\mathbf{U}_L), \quad (96)$$

$$\mathbf{F}_R = \mathbf{F}(\mathbf{U}_R), \quad (97)$$

$$\mathbf{F}_{*L} = \mathbf{F}_L + S_L(\mathbf{U}_{*L} - \mathbf{U}_L), \quad (98)$$

$$\mathbf{F}_{*R} = \mathbf{F}_R + S_R(\mathbf{U}_{*R} - \mathbf{U}_R) \quad (99)$$

$$(100)$$

with the state vector in the star region defined as

$$\mathbf{U}_{*k} = [\rho_{*k}, \rho_{*k} S_*, (\rho E)_{*k}]^T, \quad (101)$$

$$\rho_{*k} = \rho_k \frac{S_k - u_k}{S_k - S_*}, \quad (102)$$

$$(\rho E)_{*k} = \frac{(S_k - u_k)(\rho E)_k - p_k u_k + p_* S_*}{S_k - S_*}, \quad (103)$$

$$p_* = p_L + \rho_L (S_L - u_L)(S_* - u_L) \quad (104)$$

with $k \in \{L, R\}$, $S_k \in \{S_L, S_R\}$ and p_* to be identical on both sides of the contact wave.

4.2.2 Spatial Reconstruction

In order to calculate the fluxes across cell boundaries, first- and second-order spatial reconstruction schemes are implemented.

First Order In the simplest case, the flux across the cell boundary f between cells i and j is defined as

$$\mathbf{U}_{f,L} = \mathbf{U}_i, \quad \mathbf{U}_{f,R} = \mathbf{U}_j \quad (105)$$

where subscripts L and R refer to the direction from which the flux originates (cf. Figure 5).

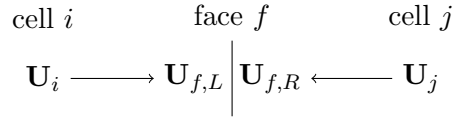


Figure 5: Flux across cell boundaries

MUSCL To reduce numerical diffusion, a second-order spatial reconstruction based on the MUSCL approach (Monotone Upstream-centered Scheme for Conservation Laws) is implemented [29]. The state within each cell is assumed to vary linearly, rather than being piecewise constant. The slopes σ_i are computed from neighboring cell values.

$$\mathbf{U}_{f,L} = \mathbf{U}_i + \frac{1}{2}\sigma_i, \quad \mathbf{U}_{f,R} = \mathbf{U}_j - \frac{1}{2}\sigma_j \quad (106)$$

To ensure stability in regions with steep gradients, slope limiters are employed [29]. Two simple limiters are considered.

minmod The minmod limiter is defined as

$$\text{minmod}(a, b) = \begin{cases} \text{sign}(a) \min(|a|, |b|), & ab > 0 \\ 0, & ab \leq 0 \end{cases} \quad (107)$$

with

$$\sigma_i = \text{minmod}(d_-, d_+) \quad (108)$$

$$d_- = \mathbf{U}_i - \mathbf{U}_{i-1} \quad (109)$$

$$d_+ = \mathbf{U}_{i+1} - \mathbf{U}_i \quad (110)$$

This limiter selects the smaller slope if both have the same sign, otherwise it suppresses the slope entirely.

Monotonized Central (MC) The monotonized central limiter extends the minmod limiter:

$$\sigma_i = \text{minmod}\left(\frac{d_- + d_+}{2}, 2d_-, 2d_+\right) \quad (111)$$

where the three-argument minmod function is defined as

$$\text{minmod}(a, b, c) = \begin{cases} \text{sign}(a) \min(|a|, |b|, |c|), & abc > 0 \\ 0, & abc \leq 0 \end{cases} \quad (112)$$

The MC limiter is used as the default limiter.

4.2.3 Time Integration

Time integration of the transport terms is performed using either a forward Euler scheme (first-order accurate) or a second-order Runge-Kutta scheme. This is embedded in a symmetric operator splitting framework with a first order source term integrator [29]. The update is given by

$$\mathbf{U}^{(1)} = \mathbf{U}^{(n)} + \frac{1}{2}\Delta t S\left(\mathbf{U}^{(n)}\right) \quad (113)$$

$$\mathbf{U}^{(2)} = \text{solve transport with } \mathbf{U}^{(1)} \quad (114)$$

$$\mathbf{U}^{(n+1)} = \mathbf{U}^{(2)} + \frac{1}{2}\Delta t S\left(\mathbf{U}^{(2)}\right) \quad (115)$$

where $\mathbf{U}^{(n)}$ is the state at the current time step, Δt is the time step size, and S represents the source terms.

Forward Euler The forward Euler scheme updates the transport step as

$$\mathbf{U}^{(2)} = \mathbf{U}^{(1)} + \Delta t L\left(\mathbf{U}^{(1)}\right) \quad (116)$$

where L denotes the operator associated with convective and viscous fluxes.

Runge-Kutta 2 To achieve second-order accuracy, the midpoint Runge-Kutta scheme is used:

$$\mathbf{U}^{(1/2)} = \mathbf{U}^{(1)} + \frac{1}{2}\Delta t L\left(\mathbf{U}^{(1)}\right) \quad (117)$$

$$\mathbf{U}^{(2)} = \mathbf{U}^{(1)} + \Delta t L\left(\mathbf{U}^{(1/2)}\right) \quad (118)$$

The RK2 scheme is used as the default integrator.

4.3 Boundary Conditions

The following explains briefly the effect of the different boundary conditions which are available in code. All boundary conditions but the periodic wall are implemented via ghost cells. To calculate the states of the cells which touch the walls, another ghost cell is projected behind the wall in order to determine the interface fluxes correctly. Dependent on the type of boundary condition, the state of the ghost cells is defined differently. Only one row of ghost cell is required with the current spatial reconstruction schemes. To define the boundary conditions appropriately, the following refers to the cell laying on the inside of the wall with subscript L and to the ghost cell with subscript G .

wall This boundary condition prescribes zero slip at the wall. Imposes a wall temperature and optionally a composition as defined by the boundary condition.

$$u_G = -u_L \quad (119)$$

$$v_G = -v_L \quad (120)$$

$$T_G = 2T_w - T_L \quad (121)$$

$$T_{V,G} = 2T_{V,w} - T_{V,L} \quad (122)$$

$$\rho_G = \rho_L \frac{T_L}{T_G} \quad (123)$$

$$\mathbf{Y}_G = \mathbf{Y}_L \quad \text{or} \quad \mathbf{Y}_G = \mathbf{Y}_{wall} \quad (124)$$

wall_euler This boundary condition assumes full tangential slip at the wall. The reflection of the momentum in normal direction to the wall is applied in primitive space.

$$u_{n,G} = -u_{n,L} \quad (125)$$

$$v_{t,G} = v_{t,L} \quad (126)$$

$$T_G = T_L \quad (127)$$

$$T_{V,G} = T_{V,L} \quad (128)$$

$$\rho_G = \rho_L \quad (129)$$

$$\mathbf{Y}_G = \mathbf{Y}_L \quad (130)$$

with subscript n and t denoting the normal and tangential velocity component relative to the wall.

reflective This boundary condition is conceptually equal to *wall_euler* with the only difference that the reflection is calculated in the conserved space and, thus, ensures loss-less conversion by skipping the conversion from and to conserved space.

wall_slip This boundary condition applies a first-order Maxwell velocity slip and temperature jump condition at the wall according to [6]. In contrast to the no-slip wall, a finite tangential velocity and temperature difference between the wall and the fluid are permitted. The slip velocity at the wall is given by

$$u_{t,w} = A \frac{2 - \sigma}{\sigma} \lambda \left. \frac{\partial u_t}{\partial n} \right|_w + \frac{3\mu}{4\rho T_{tr}} \left. \frac{\partial T_{tr}}{\partial s} \right|_w \quad (131)$$

where $A = 2/\pi$, σ is the tangential momentum accommodation coefficient, λ is the mean free path, and n and s denote the wall-normal and tangential directions, respectively.

The temperature at the wall is obtained from the temperature jump condition

$$T_w^* = T_w + \frac{2 - \alpha}{\alpha} \frac{2\gamma}{(\gamma + 1)\text{Pr}} \lambda \left. \frac{\partial T}{\partial n} \right|_w \quad (132)$$

where α is the thermal accommodation coefficient.

In the ghost cell formulation, these conditions are enforced by prescribing the ghost state such that the resulting interface values satisfy the slip relations.

$$u_{n,G} = -u_{n,L} \quad (133)$$

$$u_{t,G} = 2u_{t,w} - u_{t,L} \quad (134)$$

$$T_G = 2T_w^* - T_L \quad (135)$$

$$\rho_G = \rho_L \frac{T_L}{T_G} \quad (136)$$

$$\mathbf{Y}_G = \mathbf{Y}_L \quad \text{or} \quad \mathbf{Y}_G = \mathbf{Y}_{wall} \quad (137)$$

inflow All primitive variables are prescribed from U_{in} . The pressure is calculated from the equation of state.

$$\mathbf{U}_G = \mathbf{U}_{in} \quad (138)$$

outflow A zero-gradient condition across the cell boundary for all primitive variables.

$$\mathbf{U}_G = \mathbf{U}_L \quad (139)$$

periodic This boundary condition is not applied via ghost cells, but via the mesh connectivity.

5 Verification

This section presents the verification of the implementation according to the governing equations explained in Section 3. First, the implementation of the source terms which mainly drive the distinction between this hypersonic and a subsonic solver, is compared against results from literature in a 0D heatbath scenario in Section 5.1 and 5.2. Then, the spatial dimensionality is increased to model a shock tube scenario and compare against literature, presented in Section 5.3. At last, 2D results from a blunted cone during an atmospheric re-entry scenario are compared against reference data in Section 5.4.

5.1 0D Heatbath - Casseau

This section aims to verify the implementation of the different source terms related to hypersonic physics. The difference between this section and Section 5.2 is the reference against which the results of this work are compared against. This section compares against the results from Casseau [6].

Every verification scenario is set up with only a single cell and initialized without convective flow, but a strongly disturbed system. The equilibration behavior is compared against the reference. The different scenarios are structured so that source term contributions can be verified as isolated from another as possible. First, only the vibrational-translation temperature relaxation for a single species is verified while frozen chemistry is imposed i.e. no reactions can occur (cf. Section 5.1.1). Next, the temperature relaxation is investigated for a multispecies gas under frozen chemistry conditions (cf. Section 5.1.2). Section 5.1.3 verifies the implementation of the chemical relaxation. Finally, a chemically reacting heatbath with air is set up in Section 5.1.4.⁵

The results presented in this section can be reproduced with `experiments/heatbath_0d_casseau/heatbath_0d_casseau.ipynb`.

5.1.1 Translational-vibrational relaxation of a single-species gas

The following case compares results for a non-reacting gas of diatomic nitrogen under heating ($T^0 > T_V^0$) and cooling conditions ($T_V^0 > T^0$) with and without the consideration of the electronic energy mode. Two different internal energy models have been implemented in the scope of this work: the harmonic oscillator for the vibrational energy mode paired with the electronic level summation for the electronic energy mode according to Casseau [6] as well as the combination of both available as a curve fit proposed by Gnoffo [10] (cf. Section 3.2). The results from both thermodynamic models will be presented. Table 7 summarizes the initial conditions run for this case.

Using Bird's internal energy model as explained in Section 3.2 acceptable agreement is found in both the heating and the cooling case (cf. Fig. 6, Tab. 7, 1.1 and 1.2). The temperature traces align well with Casseau's results for a modified parameter of the relaxation time constant model to match the results from the DSMC code *dsmcFoam* [6]. The equilibrium temperature aligns perfectly in both cases confirming the correct implementation of the internal energy model including the right use of the associated datasets.

Using Gnoffo's internal energy model, similar trends can be inferred from the results (cf. Fig. 7. Though, several differences need to be highlighted from the result presented in Figure 6. Firstly, the equilibrium temperature does not match the reference. This can be explained with

⁵As the reference data is only available in form of diagrams, datapoints have been extracted manually to be able to quantitatively compare against the results of this work. Cubic splines have been used to connect these scattered data, allowing an easier comparison against continuous traces computed with this code, but being physically wrong. It is obvious that this procedure cannot guarantee perfect accuracy, but it will be good enough to extract error trends.

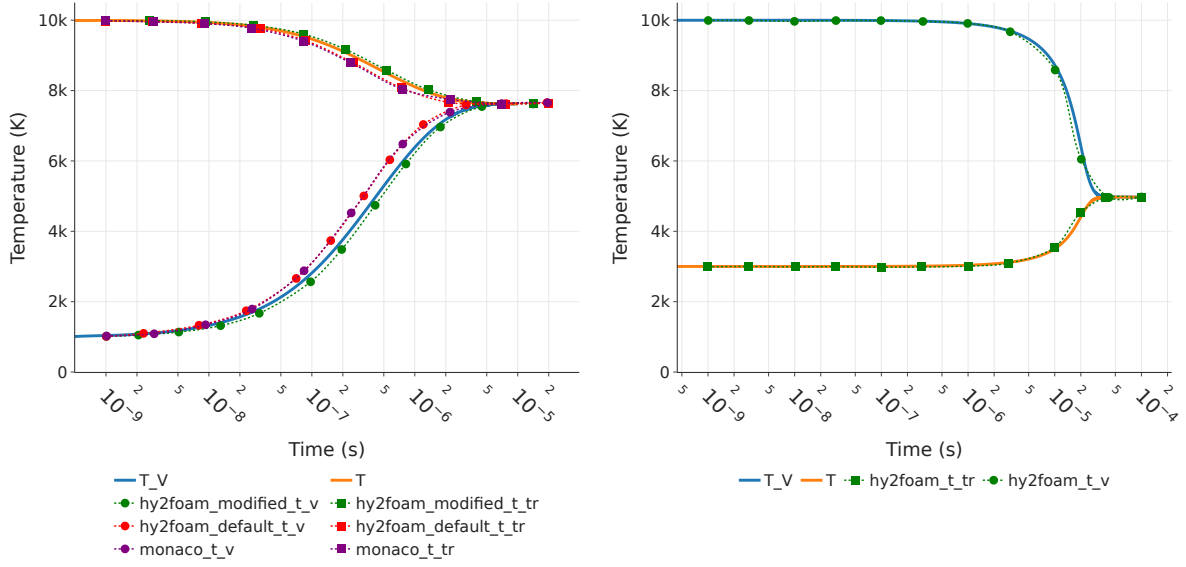


Figure 6: Thermal relaxation of an N₂ bath according to scenarios 1.1 (left) and 1.2 (right) (cf. Tab. 7). The reference indicated as "modified" uses an adapted preexponential factor to calculate τ^{MW} to match the result from a Monte-Carlo based solver while "default" refers to the baseline result using tabulated data as specified in Section 3 [6].

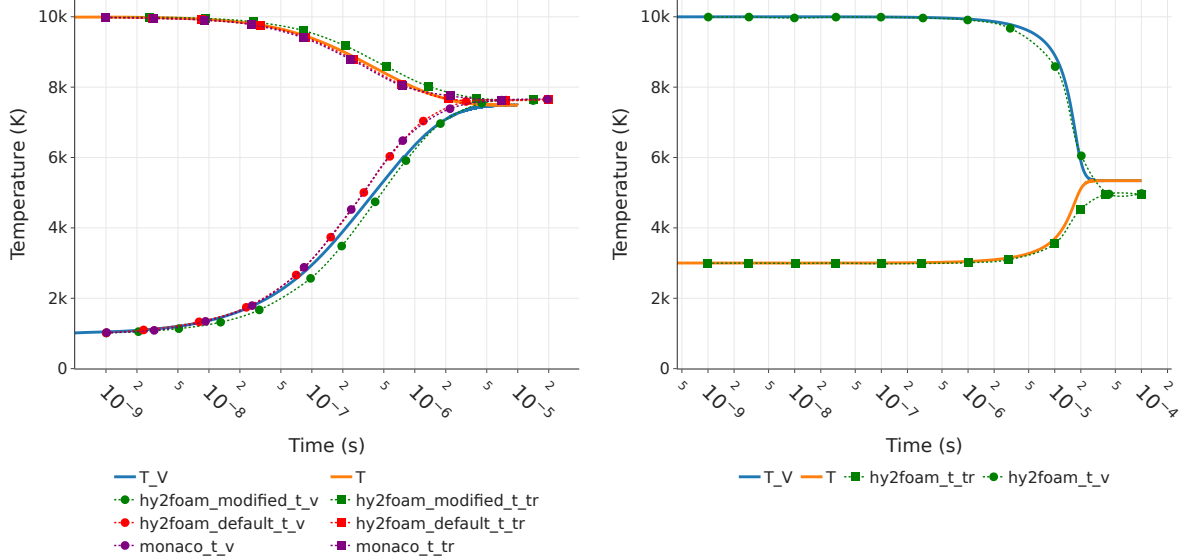


Figure 7: Thermal relaxation of an N₂ bath according to scenarios 1.3 (left) and 1.4 (right) (cf. Tab. 7). The reference indicated as "modified" uses an adapted preexponential factor to calculate τ^{MW} to match the result from a Monte-Carlo based solver while "default" refers to the baseline result using tabulated data as specified in Section 3 [6].

Case	T , [K]	T_V , [K]	p , [bar]	Vib. energy model	El. energy model	Reactions	Reference
1.1	10000	1000	1	harm. osc. [6]	None	None	[6] Fig. 3.1a
1.2	3000	10000	1	harm. osc. [6]	None	None	[6] Fig. 3.1b
1.3	10000	1000	1	curve fit [10]	curve fit [10]	None	[6] Fig. 3.1a
1.4	3000	10000	1	curve fit [10]	curve fit [10]	None	[6] Fig. 3.1b
1.5	30000	1000	1	harm. osc. [6]	el. levels [6]	None	[6] Fig. 3.2
1.6	30000	1000	1	curve fit [10]	curve fit [10]	None	[6] Fig. 3.2

Table 7: Case setups for translation-vibrational single-species relaxation

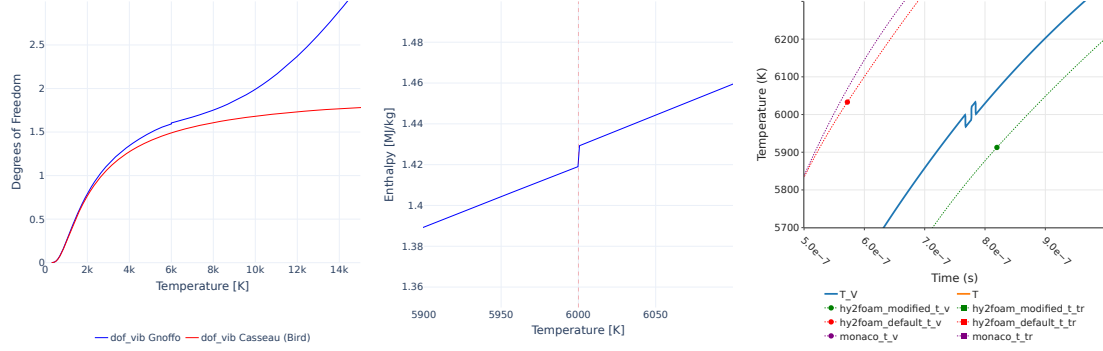


Figure 8: *Left*: Difference in number of degrees of freedom (DOF) represented by Bird’s and Gnoffo’s internal energy models. The DOF have been calculated according to Equations 49. *Center*: Jump in enthalpy curve fit proposed by Gnoffo [10]. *Right*: Unsteadiness from enthalpy jump lead to unphysical results as in the case of scenario 1.3 (cf. Tab. 7)

the curve fit which does not represent the same number of degrees of freedom (DOF) as Bird’s model with Casseau’s parameterization, as shown in Figure 8, left. At $T = 10000\text{K}$, Gnoffo’s model considers 20% more DOF than Bird’s coming from the fact the Gnoffo does already consider the electronic energy mode while the reference does not (cf. Tab. 7). More energy will be stored in the vibroelectronic energy modes using Gnoffo’s model, leading to a lower overall temperature in the heating case (Figure 7). Appendix C presents the impact on more DOF available on the equilibrium temperature in detail. To conclude, Gnoffo’s model is not useful for cases where only the vibrational energy mode shall be considered.

Another important aspect of the internal energy model proposed by Gnoffo is that the curve fit is piecewise defined and at the edges not C0-continuous according to the parameterization proposed by Gnoffo [10] (cf. Appendix B). This transfers the unsteadiness to the solver solution and ultimately leads to unphysical results (cf. Fig. 8, right). The reaction of the solver is expected: as the vibroelectronic temperature exceeds $T_V = 6000\text{K}$, more heat capacity is accessible due to this numerical inaccuracy, leading to an immediate drop in temperature so that $T < 6000\text{K}$ again.

Good agreement can also be found with the consideration of the electronic energy mode. Using Bird’s energy model, no meaningful deviation from the reference can be identified. This verifies the implementation of both the vibrational and the electronic thermodynamic models as well as the implementation of the thermal relaxation source term for a single species. The error margin shrinks if Gnoffo’s enthalpy curve fit is used. Though, there is still a mismatch in equilibrium temperature originating from a mismatch in electronic levels considered by Gnoffo to create this curve fit. This is shown in Figure 3. Similar deviations have been explained comparing the degrees of freedom predicted by the models (cf. Appendix C and as there is no logical difference in the code, the same conclusion is drawn for this case.

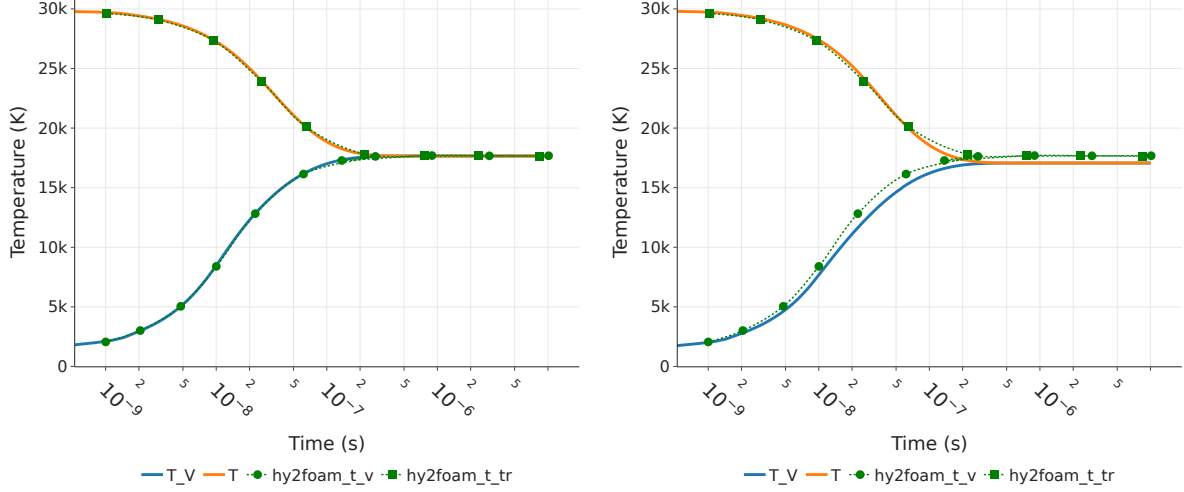


Figure 9: Thermal relaxation of an N_2 bath according to scenarios 1.5 and 1.6 (cf. Tab. 7)

To conclude, no benefit of Gnoffo's curve fit has been identified so far. The original motivation for a curve fit was most likely the reduction of species parameter look-ups to be performed per time step to increase the speed of computation. Though, this is a negligible bottleneck with the solver architecture developed in this work as the species information including electronic energy parameterization is stored in memory during the solving process.

5.1.2 Multispecies non-reactive relaxation

Now, two gas mixtures, one made of N_2 and N and the other one made of N_2 and O_2 , will be initialized with strong thermal inequilibrium at $T = 30000K$ and $T_V = 1000K$ and number densities of $n = 5e^{22}1/m^3$ for both species. The gas is imposed to be chemically frozen. This allows to check if the implementation for thermal relaxation is handling multispecies gases correctly. The comparison is only performed using the internal energy model from Bird. Figure 10 presents the results for this case. No noteworthy deviation can be identified from the reference result with which the implementation of the thermal relaxation is verified.

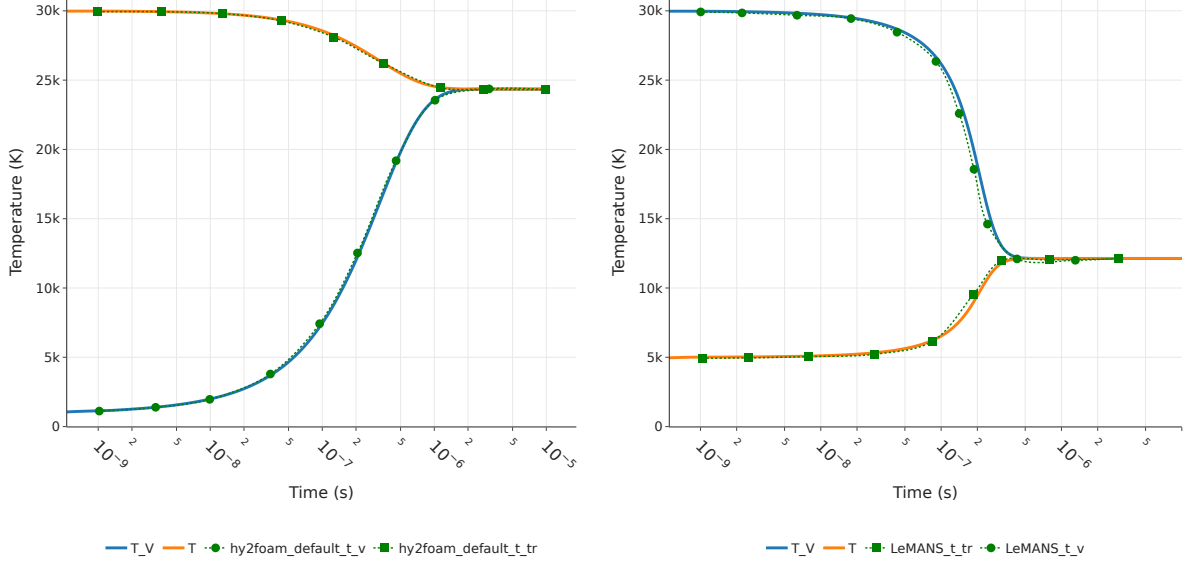


Figure 10: *Left:* Thermal relaxation of a nonreacting (N_2 , N) heatbath with reference data from [6], Fig. 3.3 *Right:* Thermal relaxation of a nonreacting (N_2 , O_2) heatbath with reference data from [6], Fig. 3.4

5.1.3 Finite rate chemistry

After the successful verification of the implementation to compute the thermal relaxation, the following compares the results for chemical relaxation to reference data. For this, only one dissociation reaction is considered



Following the same procedure as Casseau, two different Arrhenius parameter sets are used to be run with either Park's TTV model or the CVDV model (cf. Section 2.3). As this work implements all equations according to SI convention, Table 8 lists the values for the Arrhenius coefficients as used to generate the following results. The Park parameter set is used in combination with Park's preferential and nonpreferential TTV model. The quantum-kinetic (QK) parameterset proposed by Casseau is run with the CVDV model. The heatbath is initialized again with $T = 30000K$, $T_V = 1000K$ and number densities of $n = 5e^{22}1/m^3$ for both species. The results are presented in Figure 11.

Reaction Rate	$A_r, m^3/(molecule \cdot s)$	β_r	$T_{a,r}$
Park [22]	$1.162\,377 \times 10^{-8}$	-1.6	133200
Quantum-Kinetic [6]	$4.101\,53 \times 10^{-12}$	-0.62	133500

Table 8: Values for Arrhenius coefficients used to verify chemical relaxation implementation

The results using the two different Park scenarios are qualitatively very similar in both temperature and normalized number densities to the reference. The peak vibroelectronic temperature deviates by less than 1000K. Also, the peak occurs almost at the same time after start. This is considered good agreement based on Casseau's conclusion regarding their deviation to the DSMC result. For an unknown reason, the equilibration sets in $2 \times 10^{-7}s$ later than the reference. The number densities diverge almost equally for both models. A slight suppression between $t = [1 \times 10^{-7}, 2 \times 10^{-6}]s$ is noted. After $t = 2 \times 10^{-6}s$, the number density traces agree perfectly with the reference data.

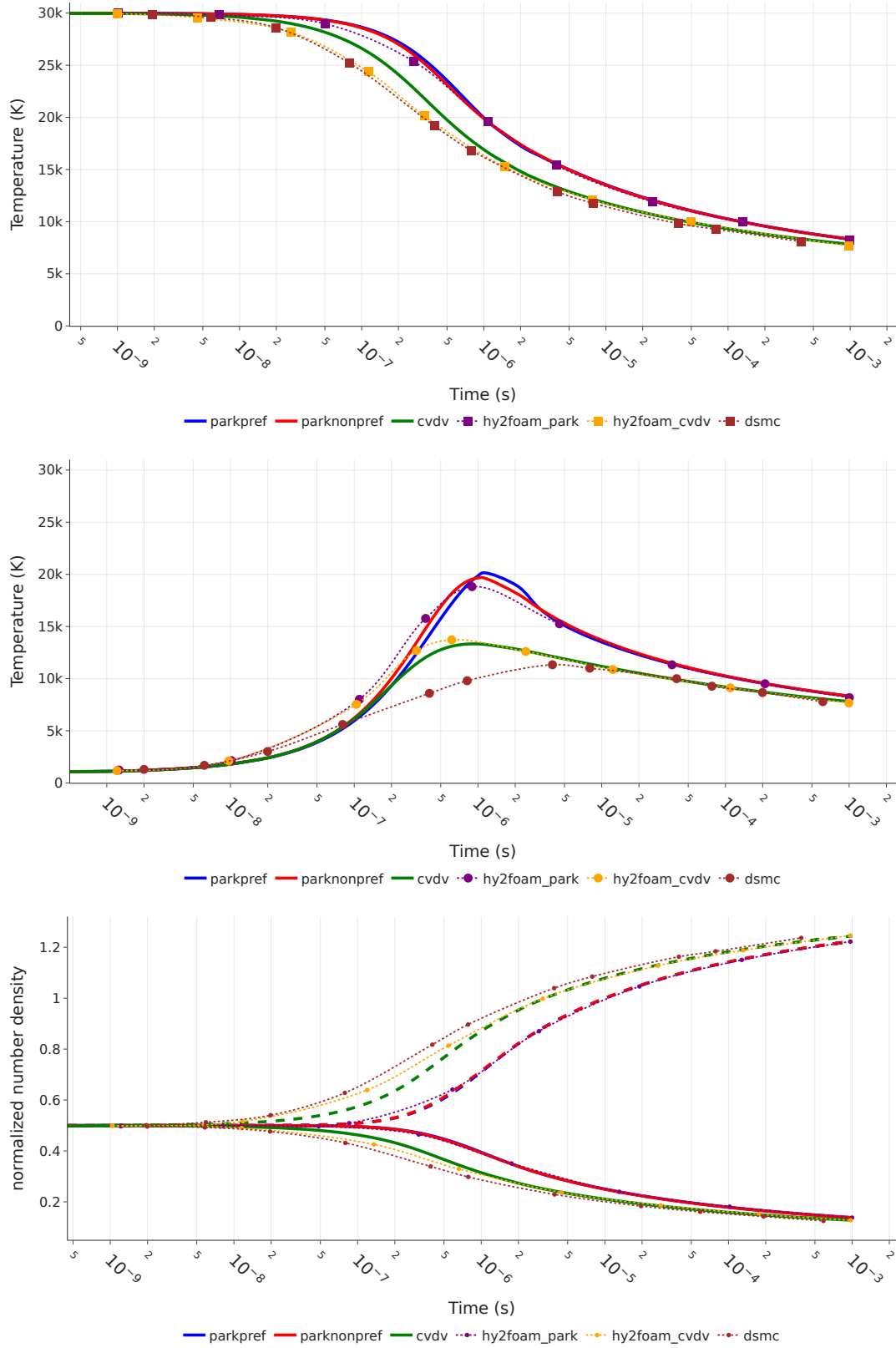


Figure 11: Chemical relaxation of a reacting (N_2 , N) heatbath with reference data from [6], Fig. 3.5. *Top*: transrotational temperature, *center*: vibroelectronic temperature, *bottom*: normalized number density

The scenario using the CVDV model with QK parameterization features a similar error trend. Again, a delay in the beginning of thermal equilibration and peak vibroelectronic temperature as well as the suppression of dissociation until about $t = 2 \times 10^{-6}$ s is visible while the peak temperature lies 400K lower than the reference.

Interestingly, both implementations suffer from a similar error. The cross section of the collisional pair could be related as different parameter values are used by literature which is discussed around Equation 56. Better agreement can be found for $\sigma_m = 3 \times 10^{-21}\text{m}^2$.

Casseau provides data for another scenario for which $T = T_V = 30000\text{K}$. It is not clear which of Park's dissociation model (preferential or nonpreferential) is used from the explanation of the case setup. A comparison of the results which are displayed in Figure 12, allow the conclusion that the nonpreferential model is configured by the reference. Due to the elevated vibroelectronic temperature, the different models to compute the forward dissociation rate k_f lead to very different results. These results have been simulated with $\sigma_m = 3 \times 10^{-21}\text{m}^2$.

The largest outlier is produced by the preferential Park model. The conclusion is obvious that the preferential model leads to a large overprediction of the vibroelectronic temperature. The transrotational temperature computed with Park's nonpreferential model aligns better with the reference. The vibroelectronic temperature lags behind in dropping by about 3×10^{-7} second. Though, the number density follows the reference well. The best results are achieved with the CVDV model. Both temperature traces and the number densities are in good agreement.

Based on the results from these two scenarios, the implementation of the CVDV model agrees best with acceptable discrepancies from the reference and is recommended to be used in further studies.

5.1.4 Air

The last verification step for which Casseau provides results, is a heatbath scenario with air at $T = T_V = 10000\text{K}$, $p = 0.063\text{atm}$ and the full reaction set of 19 reactions for the five species N_2 , N , O_2 , O and NO . The scenario is run with the CVDV model. A collision cross section of $\sigma_m = 3 \times 10^{-21}\text{m}^2$ is assumed. The results are shown in Figure 13.

All models show poor agreement. The simulation was stopped after 1×10^{-4} s as a compute time of 6 minutes were required to compute these results and an hour of additional compute time seemed not worth the extended time axis. Compared to the previous verification efforts, these deviations are large and hint to a bug in the code. Trying different reaction datasets does not achieve notably better results. At this point, the search for the bug will not be continued.

5.2 0D Heatbath - Williams

This section aims to verify the implementation of the different source terms related to hypersonic physics. The difference between this section and Section 5.2 is the simply the reference against which the results of this work are compared against. This section compares against the results from Williams et al. [30].⁶

The results presented in this section can be reproduced with `experiments/heatbath_0d_williams/heatbath_0d_williams.ipynb`.

In a similar fashion as presented by Casseau [6], Williams verifies the implementation of the HTR hypersonics CFD solver in adiabatic heatbath scenarios against results from other solvers

⁶As the reference data is only available in form of diagrams, data points have been extracted manually to be able to quantitatively compare against the results of this work. Cubic splines have been used to connect these scattered data, allowing an easier comparison against continuous traces computed with this code, but being physically wrong. It is obvious that this procedure cannot guarantee perfect accuracy, but it will be good enough to extract error trends.

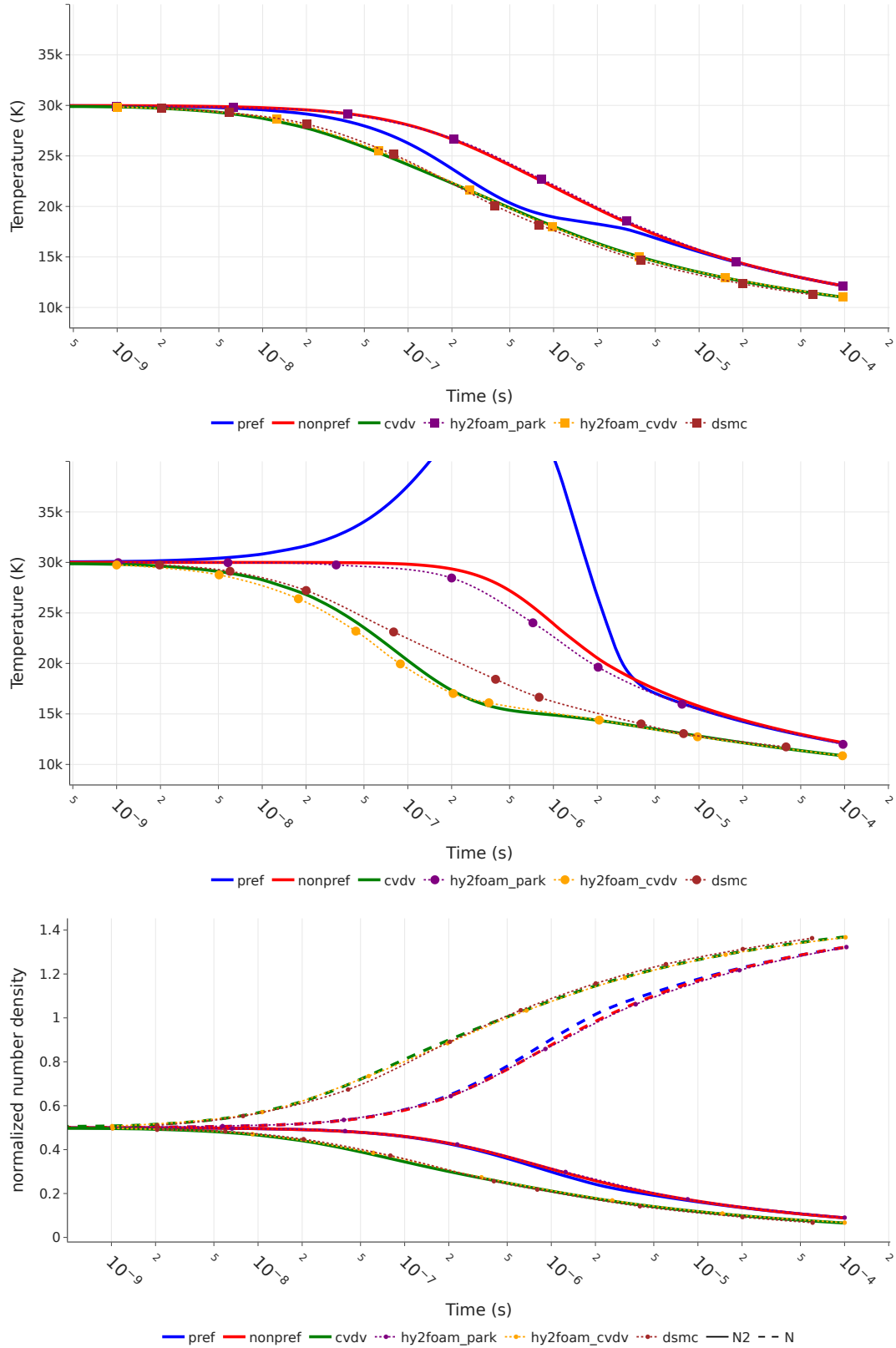


Figure 12: Chemical relaxation of a reacting (N_2 , N) heatbath with reference data from [6], Fig. 3.5

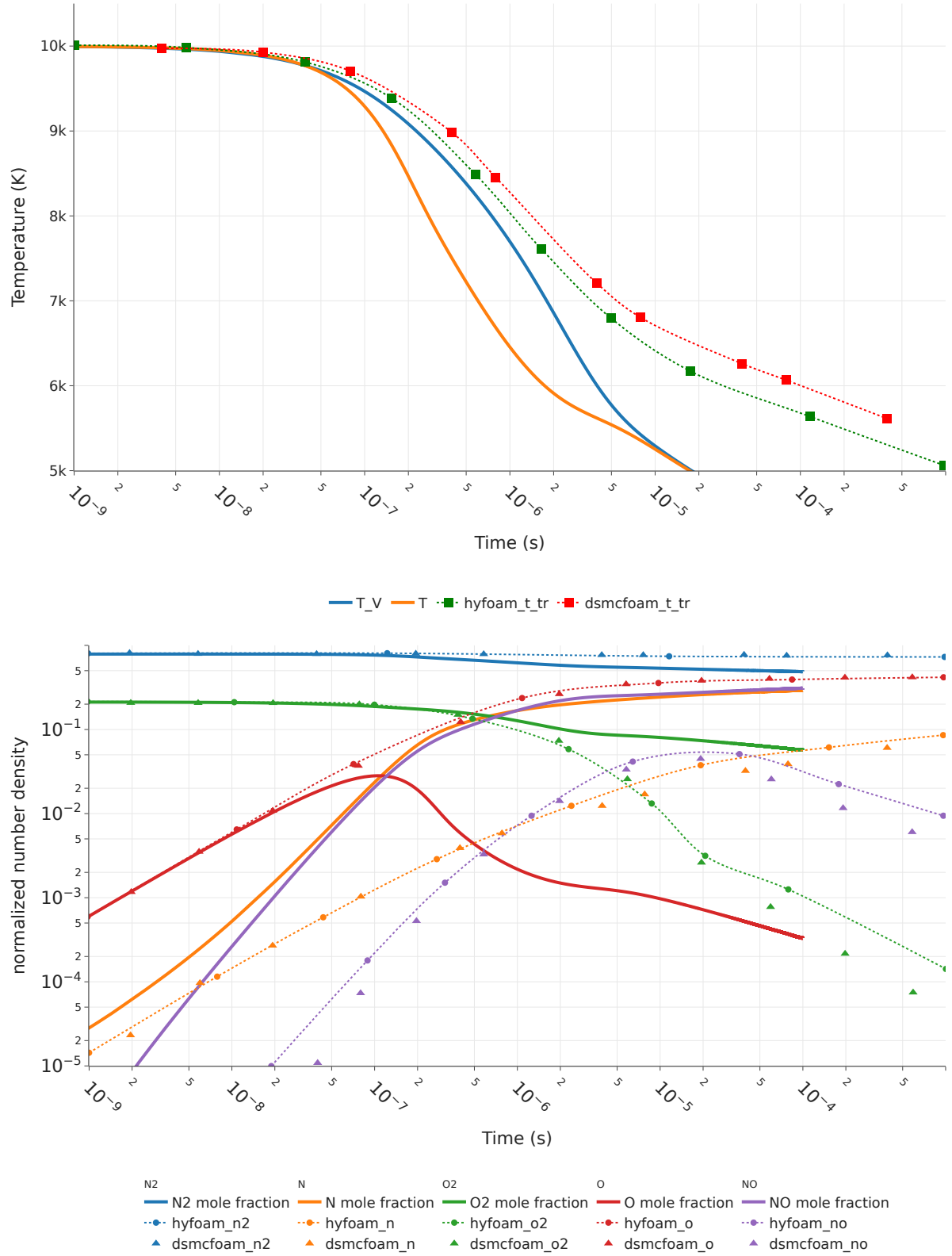


Figure 13: Chemical relaxation of a reacting heatbath with air referenced against data from [6], Fig. 3

[30]. Two cases are run: 1. air with $y_{\text{N}_2} = 0.767$, $y_{\text{O}_2} = 0.232$ at $p = 20.42\text{atm}$ with $T = 15000\text{K}$ and $T_V = 300\text{K}$, 2. pure nitrogen with $y_{\text{N}_2} = 1$, $y_{\text{N}} = 0$ at $p = 27.25\text{atm}$ with $T = 20000\text{K}$ and $T_V = 300\text{K}$. The results are presented in Figure 14 and 15.

In general, good agreement is found for the preferential and nonpreferential Park model in the case of air and pure nitrogen. For air, both models underpredict the peak temperatures of their respective reference traces by 600 K. Additionally, the equilibrium temperature is estimated too low by all models. This is a discrepancy might result from either a different parameterization of the electronic energy model, a result from different chemical reactions or from another unknown reason. For pure nitrogen, the peak temperatures by both models are overestimated. The thermal equilibration occurs faster than in the comparable case presented by Figure 11. The equilibrium temperature matches well, though. The CVDV model with QK parameterization performs the least accurate of all models. These contradictions to Section 5.1 either allude to an incorrect implementation, or to the wrong parameterization of either the thermodynamic or the chemical models or both compared to the reference data. Due to the limited reference data available, the only option seems to reproduce the results with the respective codes. The work by Williams [30] shows how good solver agreement can look like. This has not been achieved by this work. Still, at least qualitative agreement has been achieved in most cases which makes it worth to use this solver for 1D and 2D problems.

To be able to design re-entry missions with this code, one or potentially two more days invested into verification could turn out crucial for mission success.⁷

⁷This is sarcasm.

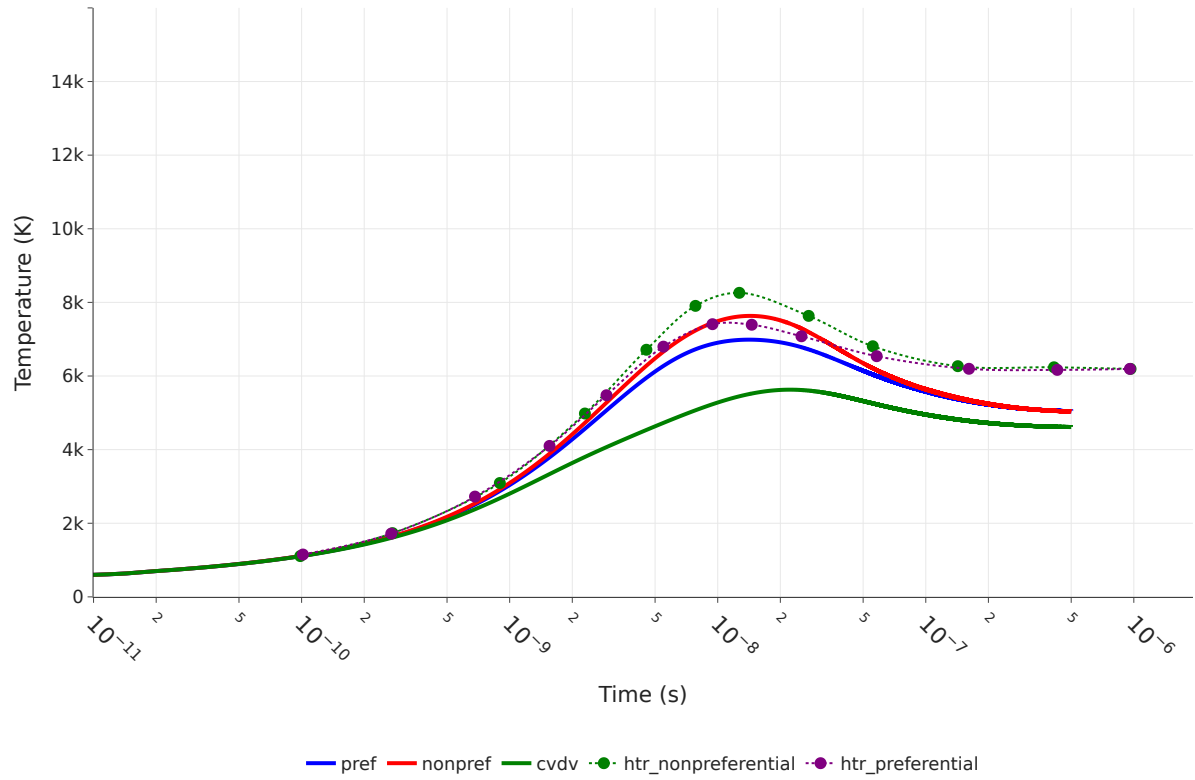
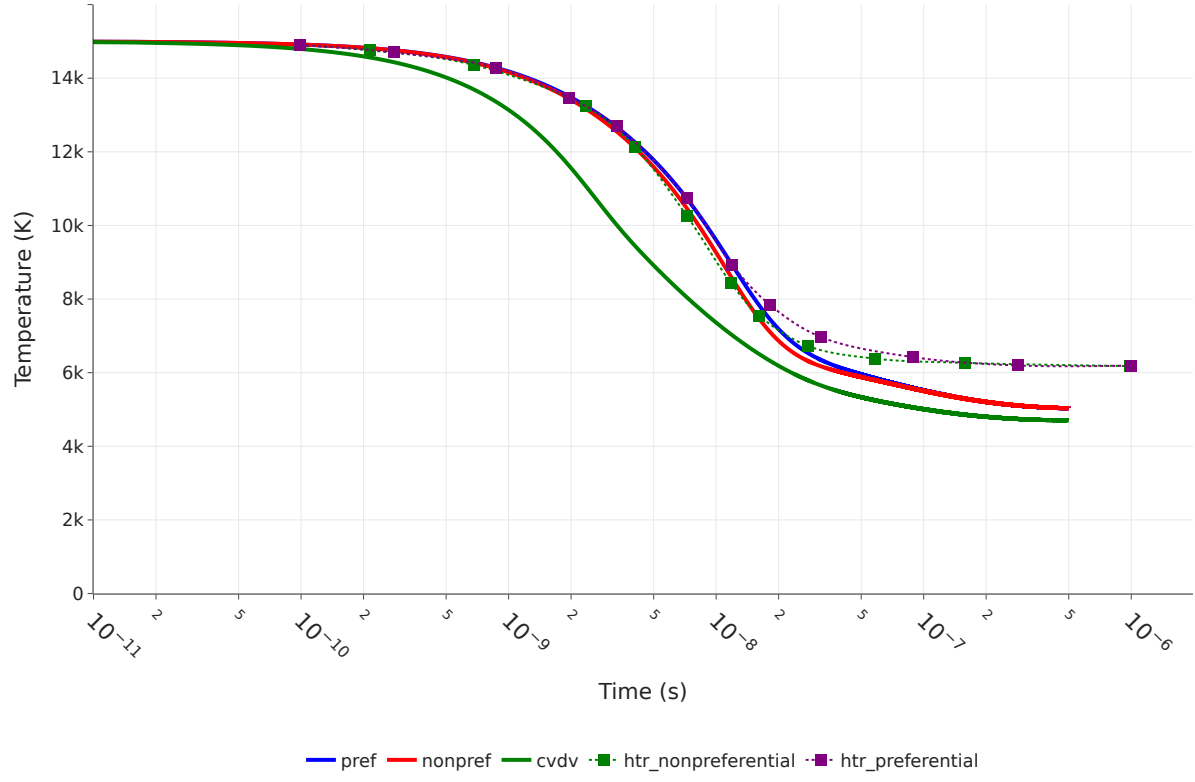


Figure 14: Chemical relaxation of a reacting heatbath with air referenced against data from [30], Fig. 1. *Top*: Transrotational temperatures, *bottom*: Vibroelectronic temperatures

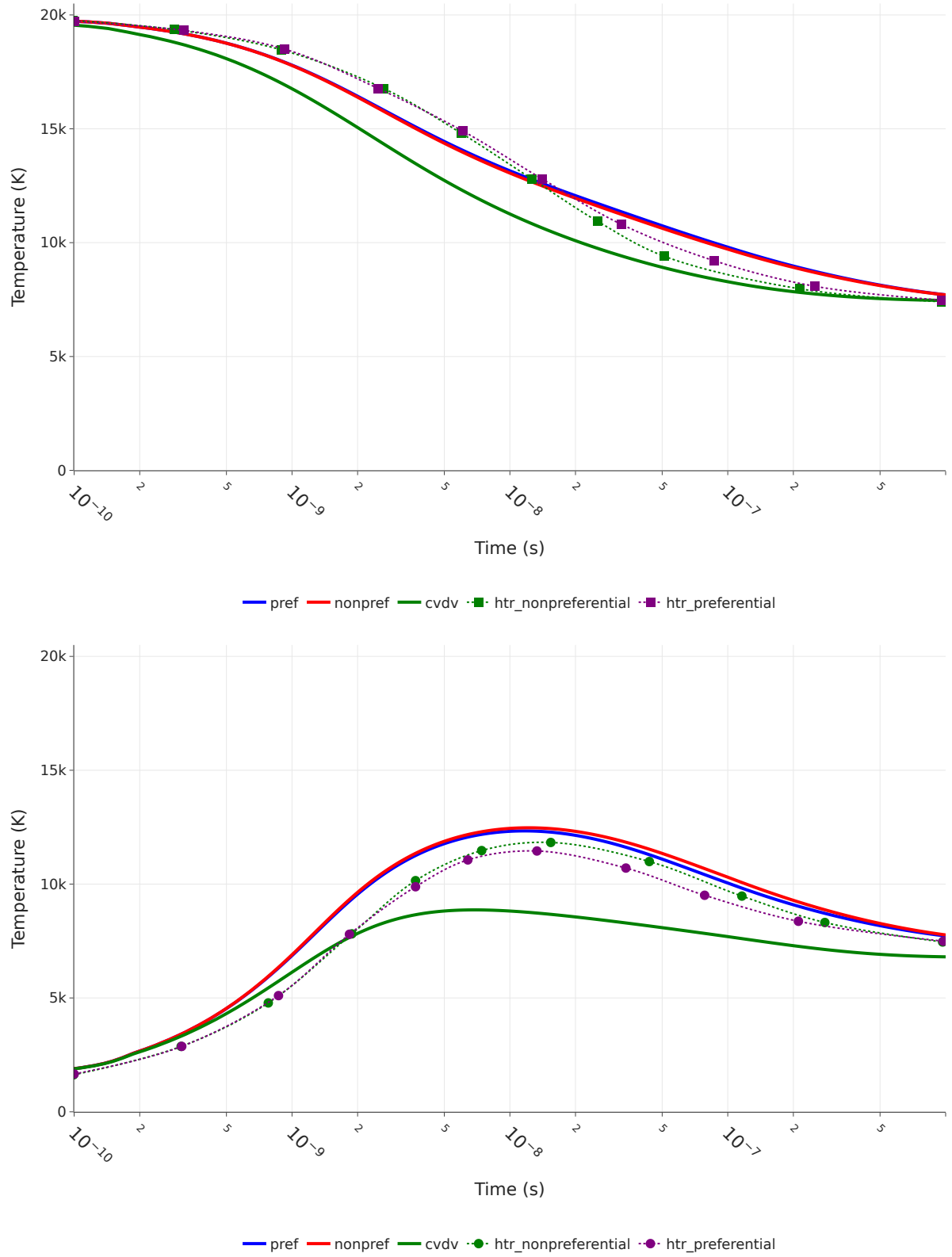


Figure 15: Chemical relaxation of a reacting heatbath with air (top) and nitrogen (bottom) referenced against data from [30], Fig. 1. *Top*: Transrotational temperatures, *bottom*: Vibro-electronic temperatures

5.3 Shock Tube

This section presents the verification effort of the solver developed in the scope of this project in a one-dimensional shock tube scenario. First, a comparison against results for a Riemann problem in subsonic conditions in Section 5.3.1 i.e. all terms related to hypersonic physics are coupled out. Section 5.3.2 then repeats this type of comparison with hypersonic contributions included and different reference data.

5.3.1 Subsonic Shock Tube

In order to evaluate the physical correctness of the convective part of the implemented HLLC solver in subsonic conditions, a shock tube scenario is set up according to Toro [29] (cf. Tab. 9). In order to compare the results against those presented by the reference, diffusive and source term contributions are simply commented out. The solver is run with a second-order Runge-Kutta time integration scheme and with textitMUSCL for second-order spatial reconstruction. The results obtained with a spatial resolution of $\Delta x = 2 \times 10^{-3}\text{m}$ and a time step width of $\Delta t = 6 \times 10^{-7}\text{s}$, are presented in Figure 16.

The results presented in this section can be reproduced with `experiments/subsonic_shock_tube_1d/shock_tube_1d.ipynb`.

ρ_L	u_L	p_L	ρ_R	u_R	p_R
1.0	0.0	1.0	0.125	0.0	0.1

Table 9: Initialization state for the shock tube scenario according to Toro [29], Table 4.1 Test 1

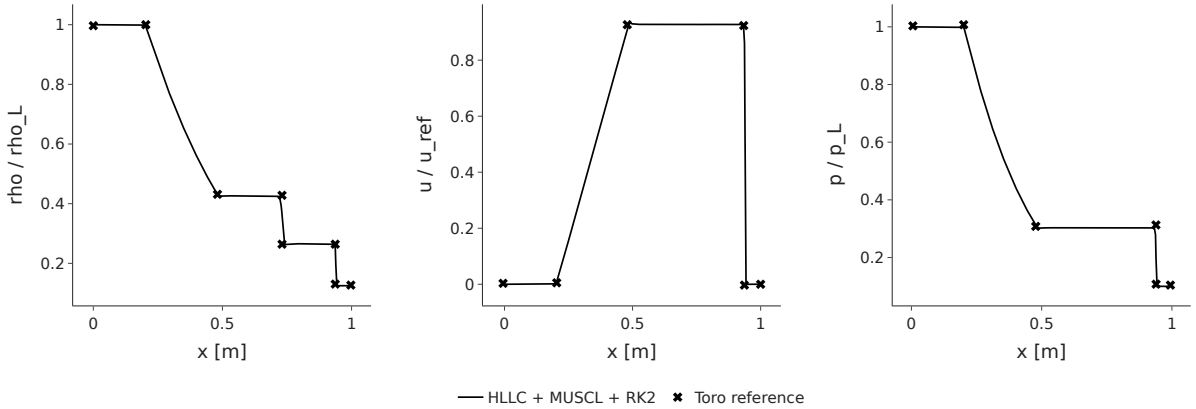


Figure 16: Results for a shock tube in subsonic conditions at $t = 0.25$ units, referenced against data from [29], Fig. 4.7

In general, good agreement can be found between the results from the implemented solver and the reference. Deviations concern the slope steepness, especially at the contact discontinuity. One contributor which was identified to add dissipation was the slope limiter of the *MUSCL* scheme. This can be reproduced by comparing the results from two identical runs with the only difference being the choice of slope limiter. For the scope of this work, the accuracy is accepted.

5.3.2 Hypersonic Shock Tube

After the successful verification of the solver in subsonic conditions, now the compound of hypersonic physics will be tested and compared against results from literature. The reference data is

extracted from Williams' verification of the HTR solver [30]. The domain is initialized according to Table 10. The solver is run with a second-order Runge-Kutta time integration scheme and with textitMUSCL for second-order spatial reconstruction. Bird's harmonic oscillator with sums over all electronic levels are combined with the Park 1990 reaction dataset [21] and the Park's preferential TTV model. The results obtained with a spatial resolution of $\Delta x = 2 \times 10^{-3}\text{m}$ and a time step width of $\Delta t = 5 \times 10^{-9}\text{s}$, are presented in Figure 16.

The results presented in this section can be reproduced with
`experiments/hypersonic_shock_tube_1d/shock_tube_1d.ipynb`.

T_L	$T_{V,L}$	p_L	gas mixture left	T_R	$T_{V,R}$	p_R	gas mixture right
12000K		2MPa	pure N ₂	1000K		100kPa	pure N ₂

Table 10: Initialization state for the shock tube scenario according to Williams [30]

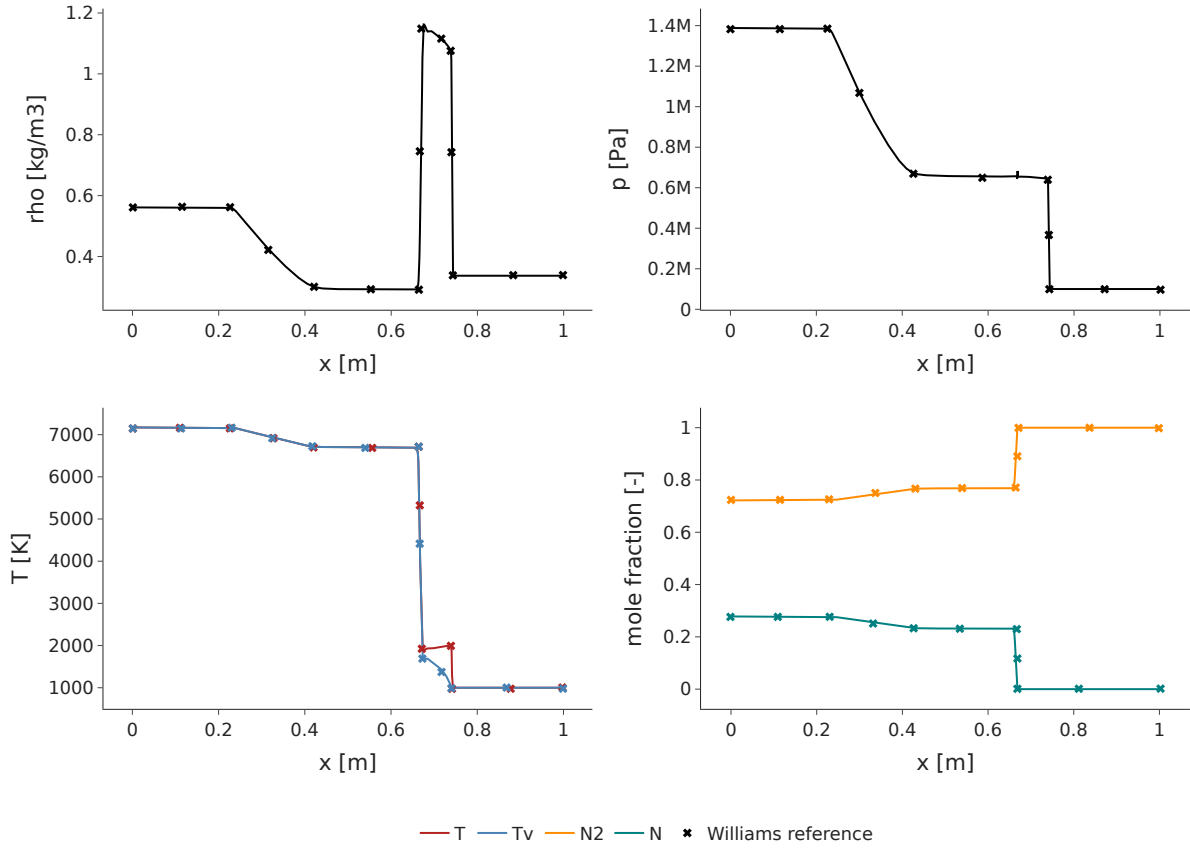


Figure 17: Results for a shock tube in hypersonic conditions at $t = 156\mu\text{s}$, referenced against data from [30], Fig. 2-5

A deviation in the slope steepness at discontinuities as already visible in the results of Section 5.3.1 can now be seen again; most pronounced in the rarefaction zone behind the shockwave at $x = 0.65\text{m}$. As this type of mismatch exists also in subsonic conditions, the conclusion is drawn that a higher order flux reconstruction is required for a better match, but that the hypersonic contributions are not connected to this. Apart from that, perfect agreement is found with the results from Williams.

To appreciate the good agreement with more advanced hypersonic solvers, Figures 18, 19 and 20 present the verified results in different time intervals. Comparing Figures 18 and 19, an understanding for the large difference in time scales can be developed. Between the first two displayed time steps in Figure 18, the complete pressure release as well as the conversion of the thermal potential into the chemical potential i.e. dissociation occurs before the shock wave has started travelling to $x \rightarrow 1\text{m}$. This is confirmed by Figure 19 which shows the first 200ns. The plot on the right of Figure 20 displays the difference between transrotational and vibroelectronic temperature over time and space. It captures the key difference of hypersonic physics to subsonic conditions well: the transrotational energy modes get excited quicker by the shockwave than the vibroelectronic modes.

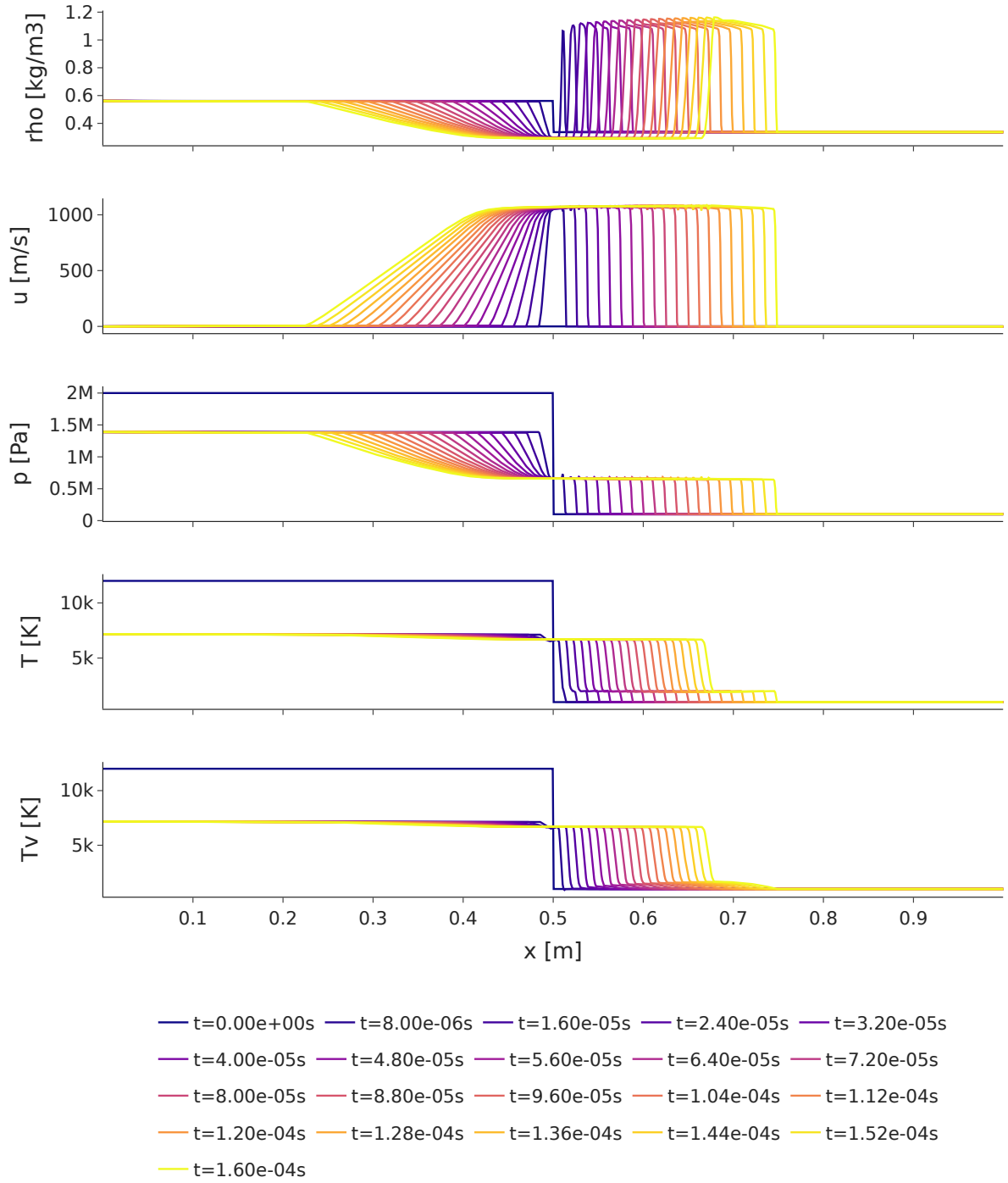


Figure 18: Transient results from a shock tube scenario in hypersonic conditions in $t = [0, 160]\mu s$

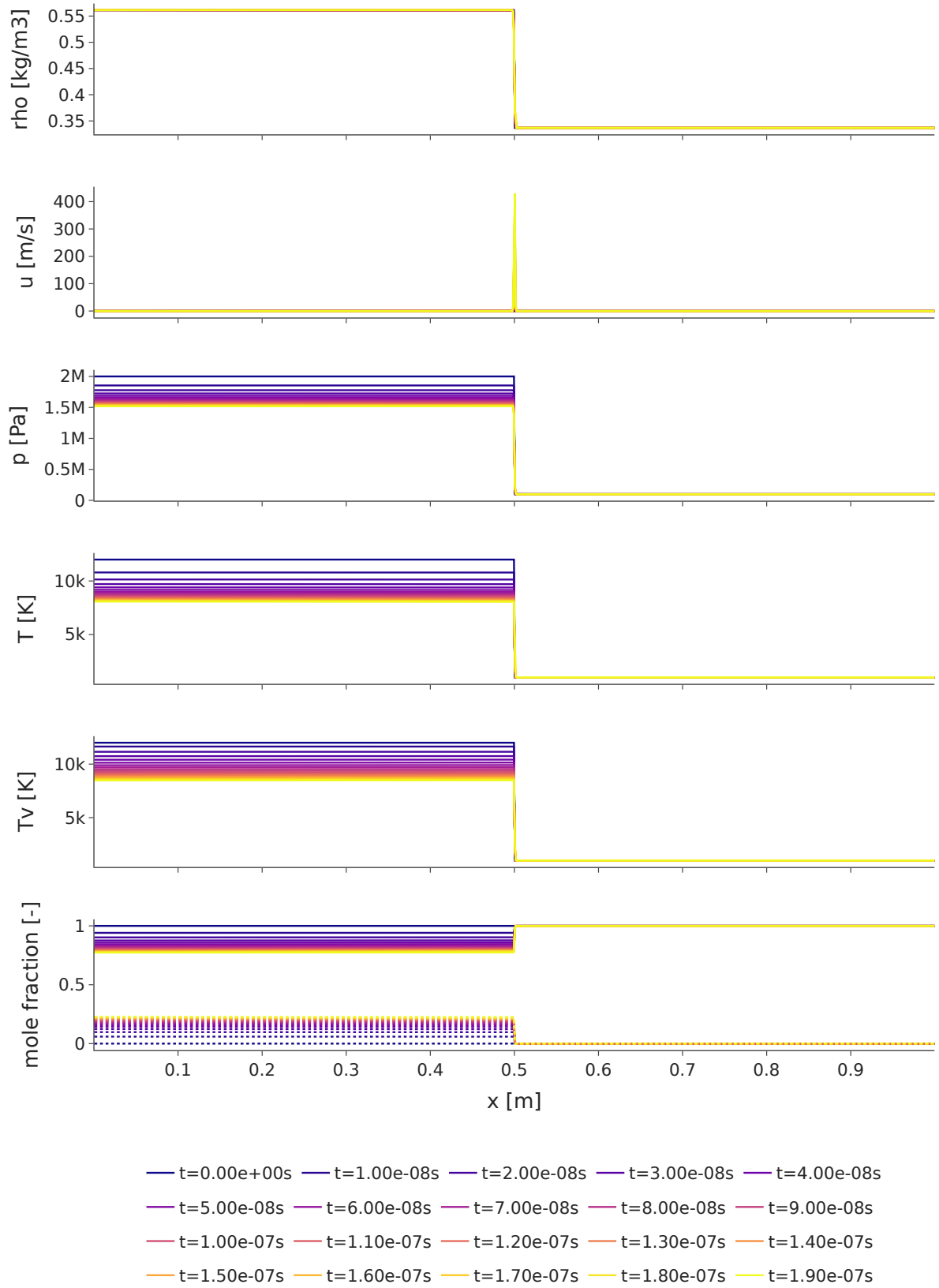


Figure 19: Transient results from a shock tube scenario in hypersonic conditions in $t = [0, 190]$ ns

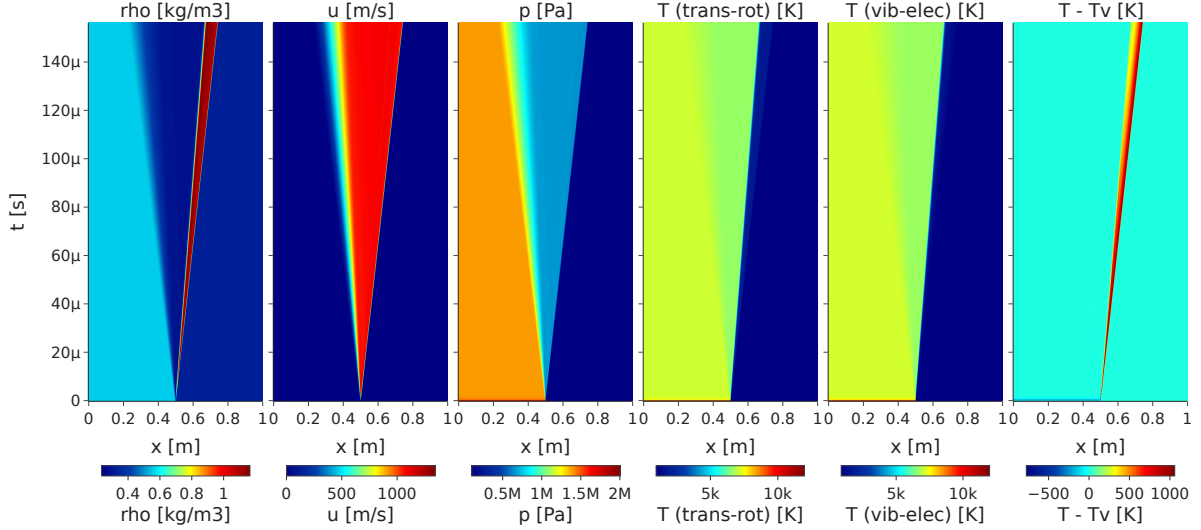


Figure 20: Transient results from a shock tube scenario in hypersonic conditions in $t = [0, 160]\mu\text{s}$

5.4 Blunted Cone

This section presents results from a chemically frozen N_2 flow at 11.3 Ma past a cone with a blunted nose. Figure 21 presents the boundaries of the mesh including their type. The initialization and parameters for the boundary conditions are listed in Table 11. This case is studied by literature and allows a comparison of the 2D implementation of the solver. Important to mention is that literature computes on a 2D axisymmetric mesh. In the scope of this work, an axisymmetrical boundary condition has not been implemented. Instead, the solver is simplifying the problem to strictly 2D. This needs to be taken into account in any interpretation. Casseau plots the primitive variables along the stagnation line which will be used as reference [6].

The results presented in this section can be reproduced with `experiments/blunt_cone_2d/run_blunt_cone.py`.

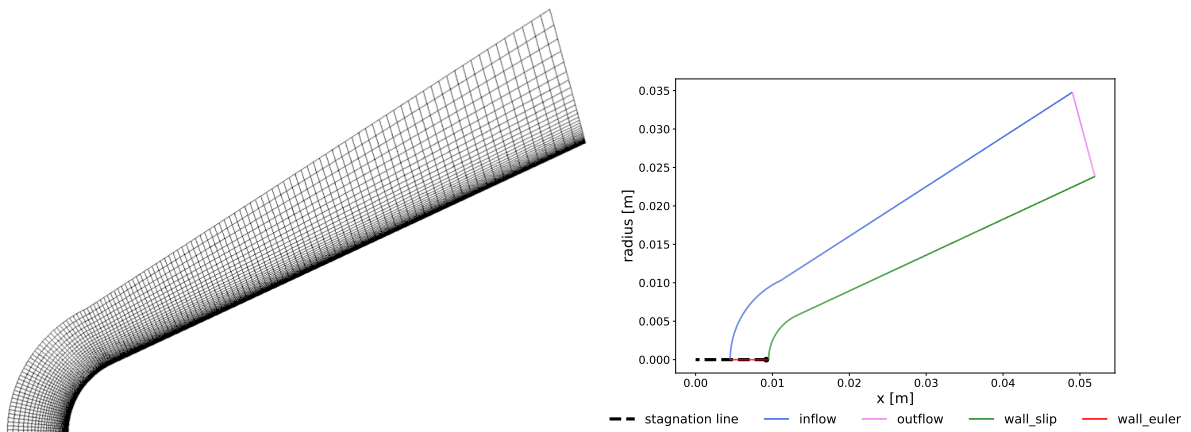


Figure 21: *Left*: mesh for blunted cone, figure taken from [6] *right*: mesh boundaries of blunted cone

The mesh published on the *hy2foam* repository contains 600 nodes in normal direction and 200 in tangential [31]. On a *g2-standard-8* machine with an *NVIDIA L4* accelerator it takes about 2hr to solve the first $10\mu\text{s}$ with a fixed temporal resolution of $\Delta t = 10\text{ps}$. Larger time steps

Free-stream velocity, U_∞	2764.5 m/s
Free-stream pressure, p_∞	21.9139 Pa
Free-stream density, ρ_∞	5.113×10^{-4} kg/m ³
Free-stream temperature, T_∞	144.4 K
Free-stream mean-free-path, λ_∞	1.01×10^{-4} m
Overall Knudsen number, Kn_{ov}	0.002
Wall temperature, T_w	297.2 K

Table 11: Initialization and boundary conditions

could be explored, though, instabilities were found with a $\Delta t = 1$ ns.

Results are shown in Figure 22. The formation of a shock layer is well visible. Right in front of the nose cone a stagnation point develops in which the velocity is zero. Looking at the difference in transrotational and vibroelectronic temperature, the time lag in the equilibration develops as expected. Figure 23 presents the normalized primitive variables along the stagnation line which is drawn schematically in Figure 21, right. The more opaque lines in each plot visualize the result at every 1 μ s. This allows to infer on the stage of convergence. General similarity can be found between the converged results and the reference data. An obvious difference is the shocklayer thickness. While Casseau’s results predicts a shock layer thickness of roughly $t = 1.1$ mm, the code of this project estimates a layer thickness of about $t = 2$ mm. This mismatch is related to the fact that this code solves only a 2D instead of a 2D axisymmetric mesh. If the results were normalized over the shock thickness layer, the temperature would match well. Similar is expected from the density and velocity distribution. While the density profile is flattened out due to the thicker shock layer, Casseau’s result estimates a higher density within the layer. The integral of the density profiles below both curves match up well enough to assume that this would be resolved by a 2D axisymmetric setup. The velocity profile within the shock layer features the same slope steepnesses with the difference that the profile extends further away from the vehicle.

With this the verification efforts within the scope of this semester thesis are finished. Good agreement with different, well-established hypersonic CFD solvers in specific verification scenarios is shown for inert single and multi-species cases for different dissociation and thermodynamic models. Perfect agreement could not be achieved. Being able to generate individual reference data with solvers which are used as benchmarks, would allow running an even larger variety of benchmark tests to understand the deviation characteristics and ultimately trace down the responsible code sections. Not even qualitative agreement can be achieved in the case of chemically reacting air proposed by Casseau. Increasing dimensionality to 1D via a Riemann problem at hypersonic conditions shows good quantitative agreement. No obvious deviation can be observed. A 2D case was run, though, the reference data was generated with a 2D axisymmetric setup while the solver of this project is constrained to 2D cases only. Thus, conclusions are limited.

Primitive variables — $t = 1.00\text{e} - 05 \text{ s}$

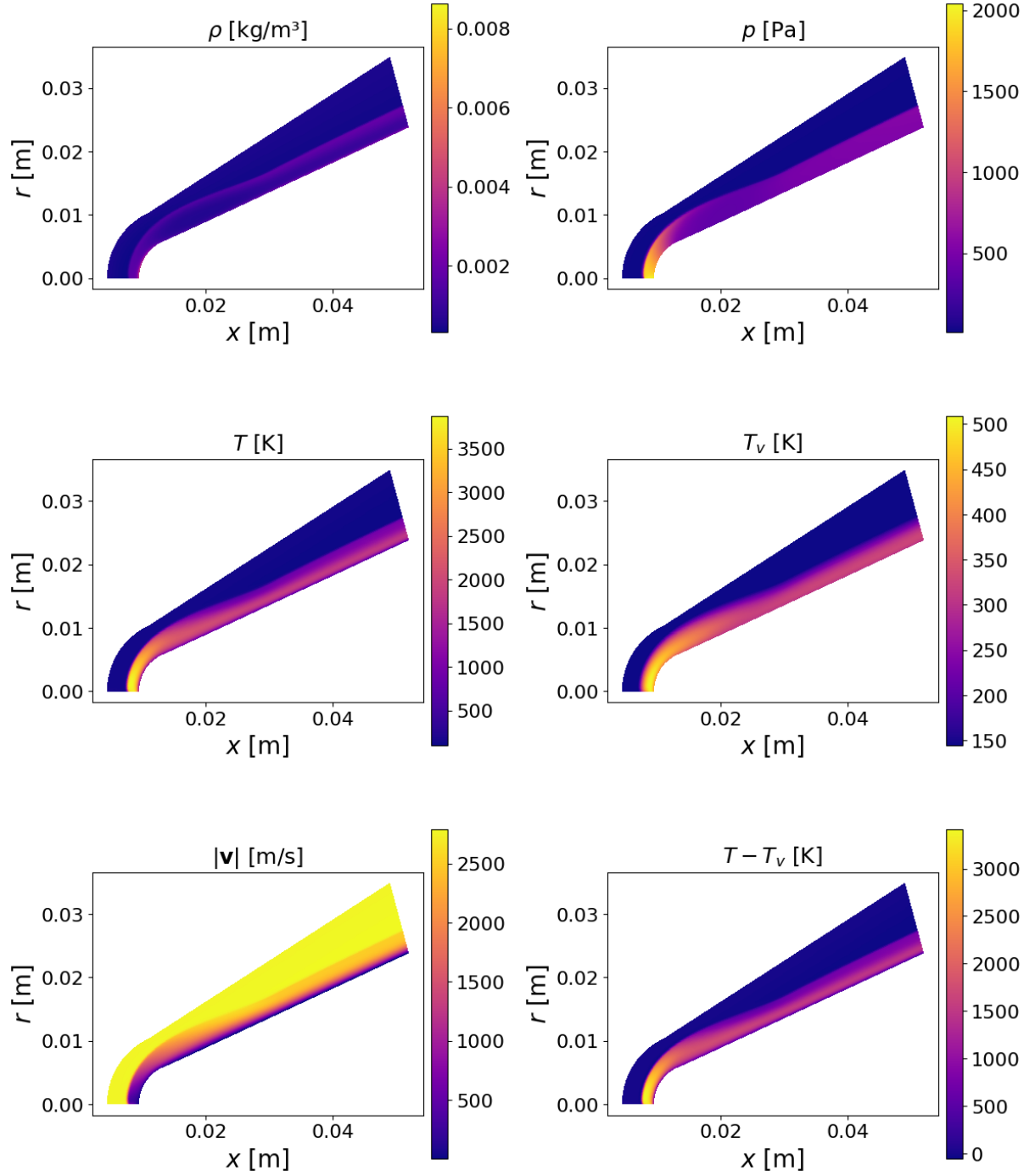


Figure 22: Primitives after convergence for initialization according to Table 11

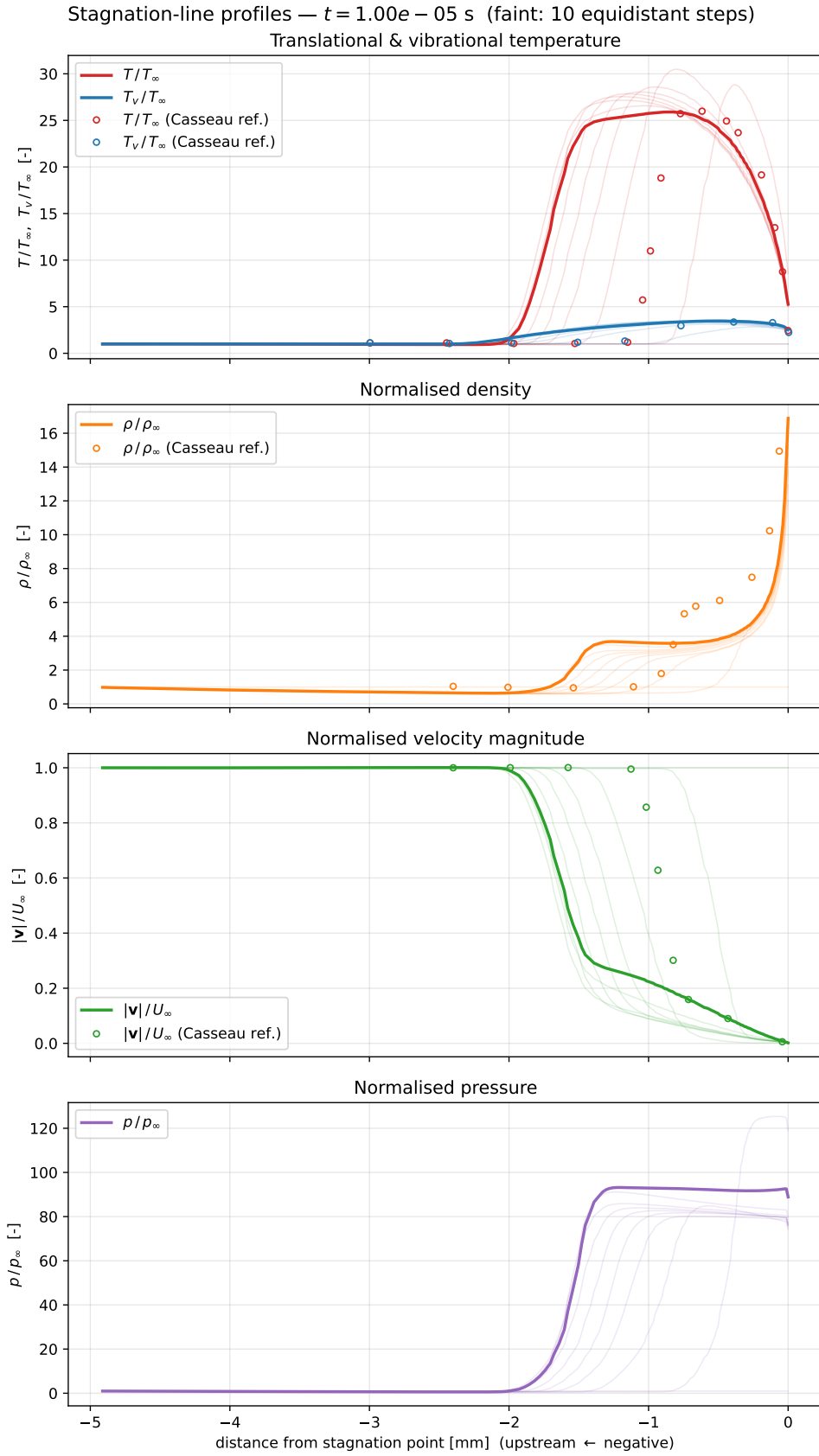


Figure 23: Primitives along stagnation line in comparison with reference data from [6]

6 Future Work

This work aimed to build a CFD solver capable of capturing hypersonic effects. A series of verification efforts have been run with mixed results. Good qualitative agreement could be shown in most cases. Though, disagreement was found in other cases. Before any extension of the solver is considered, confidence must be established regarding the performance in isolated scenarios like the heatbaths shown. Also, no dedicated verification of the transport implementation was shown. The 1D shock tube scenario can be considered as an integration test which also covers transport. Nevertheless, a dedicated verification is outstanding.

Adding ionization physics is an interesting next step which allows to achieve more realistic physical representation. The core literature referenced in this work from Gnoffo [10], Casseau [6] or Scalabrin [19] presents approaches to integrate these physics into the solver. With confidence in the solver performance at low dimensionalities, expanding to 3D or 2D axisymmetric offers to compare against more advanced metrics and study their behavior along the stagnation line through the shock wave.

A Appendix

Reaction	A · casseau_qk	A · park	A · scanlon	A · park1990	beta · casseau_qk	beta · park	beta · scanlon	beta · park1990	E_f/k
O2 + N -> O + O + N	2.28e-11	4.816e-07	1.1e-10	1.661e-08	-0.85	-2	-1	-1.5	5.937e+04
O2 + NO -> O + O + NO	3.45e-11	1.607e-07	1.1e-10	3.321e-09	-0.88	-2	-1	-1.5	5.937e+04
O2 + N2 -> O + O + N2	6.66e-11	1.607e-07	1.3e-10	3.321e-09	-0.93	-2	-1	-1.5	5.937e+04
O2 + O2 -> O + O + O2	5.91e-11	1.607e-07	5.33e-11	3.321e-09	-0.96	-2	-1	-1.5	5.937e+04
O2 + O -> O + O + O	3.08e-11	4.816e-07	1.5e-10	1.661e-08	-0.88	-2	-1.05	-1.5	5.937e+04
N2 + O -> N + N + O	5.74e-10	8.269e-08	4e-12	4.982e-08	-1.08	-1.6	-0.54	-1.6	1.135e+05
N2 + O2 -> N + N + O2	5.53e-10	6.144e-09	1.5e-11	1.162e-08	-1.06	-1.6	-0.68	-1.6	1.135e+05
N2 + NO -> N + N + NO	8.07e-10	8.269e-09	1.5e-11	1.162e-08	-1.1	-1.6	-0.68	-1.6	1.135e+05
N2 + N2 -> N + N + N2	2.19e-09	6.144e-09	4.1e-12	1.162e-08	-1.21	-1.6	-0.62	-1.6	1.135e+05
N2 + N -> N + N + N	6.69e-10	2.657e-08	1e-11	4.982e-08	-1.09	-1.6	-0.68	-1.6	1.135e+05
NO + N2 -> N + O + N2	1.75e-10	1.32e-06	2.1e-10	8.303e-15	-1	-2	-1	0	7.554e+04
NO + O2 -> N + O + O2	1.45e-10	1.32e-06	2e-10	8.303e-15	-0.99	-2	-1	0	7.554e+04
NO + NO -> N + O + NO	2.27e-10	1.32e-06	1e-10	1.827e-13	-1.06	-2	-1	0	7.554e+04
NO + O -> N + O + O	1.39e-10	1.32e-06	4e-10	1.827e-13	-1	-2	-1.1	0	7.554e+04
NO + N -> N + O + N	1.29e-10	1.32e-06	4e-10	1.827e-13	-0.99	-2	-1.1	0	7.554e+04
NO + O -> O2 + N	1.45e-20	1.39e-17	2.3e-19	1.395e-17	0.67	0	0.5	0	1.969e+04
N2 + O -> NO + N	1.93e-13	1.069e-12	8e-17	1.063e-12	-0.79	-1	0	-1	3.748e+04
O2 + N -> NO + O	4.6e-17	1.39e-17	4e-15	1.395e-17	0.04	0	-0.39	0	1449
NO + N -> N2 + O	4.82e-15	1.069e-12	5e-16	1.063e-12	-0.53	-1	-0.35	-1	1449

Figure 24: Comparison of different Arrhenius parameter sets (in the same order: [6], [21], [22], [19]). Units: $C_f \rightarrow [\text{m}^3/(\text{molecule s})]$, $n_f \rightarrow [-]$, $E/k \rightarrow [\text{K}]$

B C0-Discontinuity of Enthalpy Curve Fits

The following presents the C0-discontinuity of the curve fits proposed by Gnoffo [10] to determine species enthalpies. This is demonstrated with an example with N_2 at $T = 6000K$ which corresponds to the edge of the piecewise defined enthalpy function. The vibroelectronic enthalpy is computed according to Equation 39. The coefficients from Table 2 in [10] are listed again in Table 12.

Coefficient	$1000 \leq T \leq 6000 \text{ K}$	$6000 \leq T \leq 15000 \text{ K}$
A_1	2.8963190×10^0	3.7270000×10^0
A_2	1.5154860×10^{-3}	4.6840000×10^{-4}
A_3	$-5.7235270 \times 10^{-7}$	$-1.1400000 \times 10^{-7}$
A_4	$9.9807390 \times 10^{-11}$	$1.1540000 \times 10^{-11}$
A_5	$-6.5223550 \times 10^{-15}$	$-3.2930000 \times 10^{-16}$
A_6	-9.0586200×10^3	-1.0430000×10^4

Table 12: Enthalpy coefficients for N_2 from Table 2 [10]

Inserting the coefficients from Table 12 into Equation 39

$$\begin{aligned}
 h_{ve}^{(1000-6000)}(T) = R_{N_2} & \left(2.896319 T + \frac{1.515486 \times 10^{-3}}{2} T^2 - \frac{5.723527 \times 10^{-7}}{3} T^3 \right. \\
 & + \frac{9.980739 \times 10^{-11}}{4} T^4 - \frac{6.522355 \times 10^{-15}}{5} T^5 - 9.058620 \times 10^3 \Big) \\
 & - \frac{7}{2} R_{N_2} (T - T_{ref})
 \end{aligned}$$

$$\begin{aligned}
 h_{ve}^{(6000-15000)}(T) = R_{N_2} & \left(3.727 T + \frac{4.684 \times 10^{-4}}{2} T^2 - \frac{1.140 \times 10^{-7}}{3} T^3 \right. \\
 & + \frac{1.154 \times 10^{-11}}{4} T^4 - \frac{3.293 \times 10^{-16}}{5} T^5 - 1.043 \times 10^4 \Big) \\
 & - \frac{7}{2} R_{N_2} (T - T_{ref})
 \end{aligned}$$

leads to

$$\begin{aligned}
 h_{ve}^{(1000-6000)}(6000) &= 1419020 \text{ J/kg}, \\
 h_{ve}^{(6000-15000)}(6000) &= 1428996 \text{ J/kg}
 \end{aligned}$$

once evaluated with $T_{ref} = 298.16K$ [10]. Hence, a discontinuity of approximately $\Delta h_{ve} \approx 10000 \text{ J/kg}$ is present at the boundary between the validity intervals for this example. This discontinuity arises directly from the curve fits and is therefore not introduced by the present implementation.

This exciting numerical demonstration is found in

`jaxfluids_internship/experiments/debug/enthalpy_curve_fits.ipynb`.

C Equilibrium Temperature from Energy Conservation

This section addresses deviations seen in the equilibrium temperature T^{eq} of thermal relaxation cases using Gnoffo's enthalpy model. The results obtained and presented in Figure 6 and 7 are confirmed. The approach taken for this bases on conservation of total energy during the relaxation process.

$$\zeta_{tot}^{eq} T_{ov}^{eq} = \zeta_{tot}^{(0)} T_{ov}^{(0)} \quad (141)$$

where T_{ov} refers to the overall temperature according to Equation 46, superscripts 0 and eq denoting initial and equilibrium temperature and ζ_{tot} as the total number of degree of freedom (DOF) with

$$\zeta_{tot} = \sum_q \zeta_q \quad . \quad (142)$$

To confirm the results from Section 5.1, the following example assumes N_2 . The heating and cooling scenario of Table 7 will be analyzed for both internal energy models.

Heating With $T^{(0)} = 10000K$ and $T_V^{(0)} = 1000K$, initial overall temperature and initial DOF are evaluated as $T_{ov,Bird}^{(0)} = 9588K$ and $\zeta_{tot,Bird}^{(0)} = 5.2399$ for Bird's model and $T_{ov,Gnoffo}^{(0)} = 9572.7K$ and $\zeta_{tot,Gnoffo}^{(0)} = 5.2492$ for Gnoffo's model using the curve fit. Equation 141 must be solved implicitly for T^{eq} as ζ_{tot}^{eq} depends on T^{eq} , too. Using Newton iterations, $T_{Bird}^{eq} = 7623.3K$ and $T_{Gnoffo}^{eq} = 7490K$ with $\zeta_{tot,Bird}^{eq} = 6.5903$ and $\zeta_{tot,Gnoffo}^{eq} = 6.7088$ are obtained. These equilibrium temperatures match the results from Figure 6.

Cooling The same strategy is employed for $T^{(0)} = 3000K$ and $T_V^{(0)} = 10000K$. Initial overall temperature and initial DOF are evaluated as $T_{ov,Bird}^{(0)} = 4762K$ and $\zeta_{tot,Bird}^{(0)} = 6.6818$ for Bird's model and $T_{ov,Gnoffo}^{(0)} = 4993.5K$ and $\zeta_{tot,Gnoffo}^{(0)} = 6.9909$ for Gnoffo's model. Using Newton iterations, $T_{Bird}^{eq} = 4973K$ and $T_{Gnoffo}^{eq} = 5342.60K$ with $\zeta_{tot,Bird}^{eq} = 6.3981$ and $\zeta_{tot,Gnoffo}^{eq} = 6.5341$ are obtained. These equilibrium temperatures match the results from Figure 6. Refer to `jaxfluids_internship/experiments/debug/gnoffo_dof.ipynb` to reproduce the results.

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